



TARGET-90

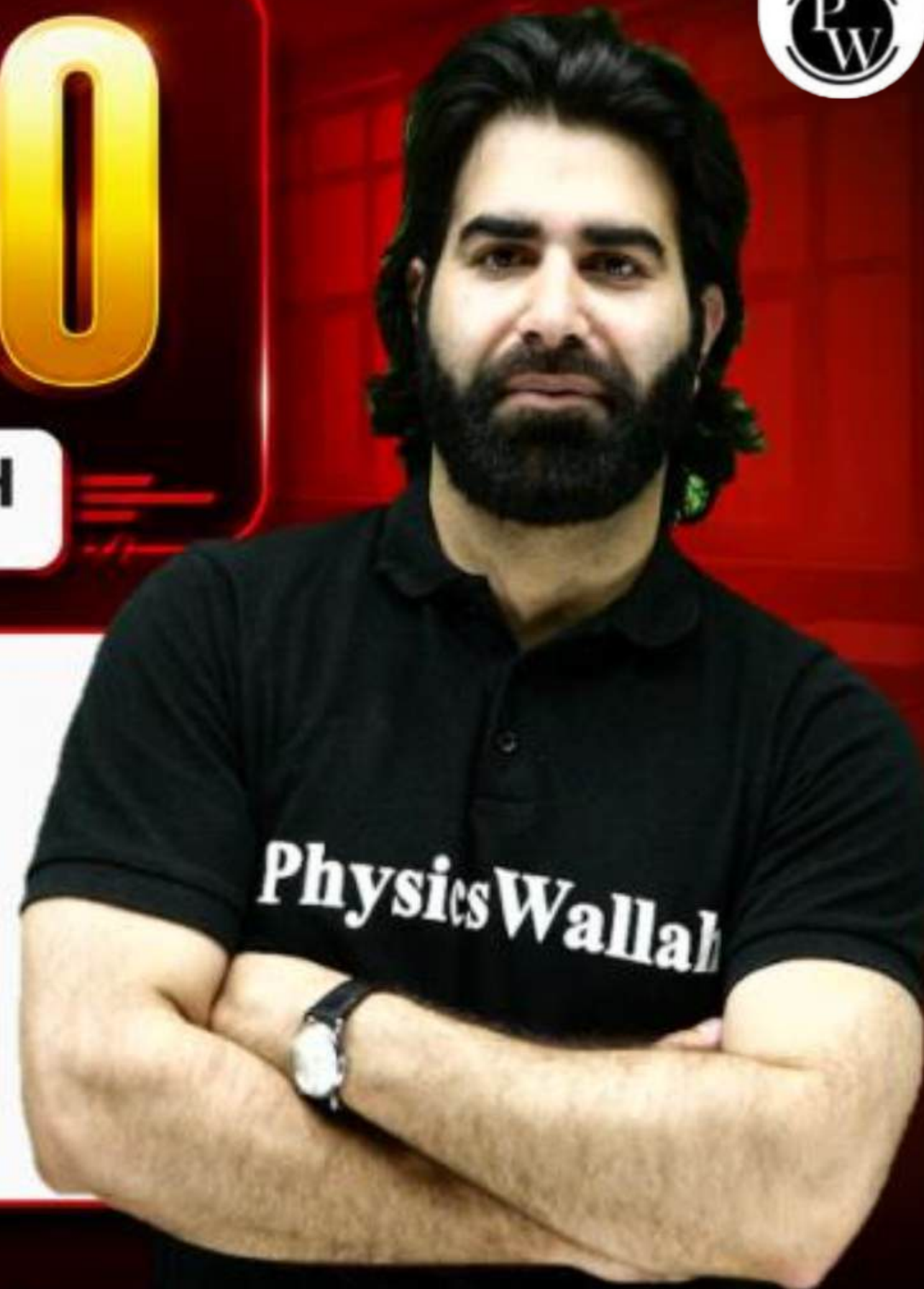
JKBOSE CLASS-11TH

Biomolecules

One Shot

Zoology

Dr Adil Altaf Sir



Topics *to be covered*



- 1 Biomolecules ✓
- 2 CARBOHYDRATES ✓
- 3 PROTEIN ✓
Lipids ✓
- 4 Nucleic Acid ✓
Enzymes ✓

Remaining Portion of Frog

Frog - Reproduction

→ Sexual Reproduction

→ Fertilization = External (Water)

→ Development = Indirect

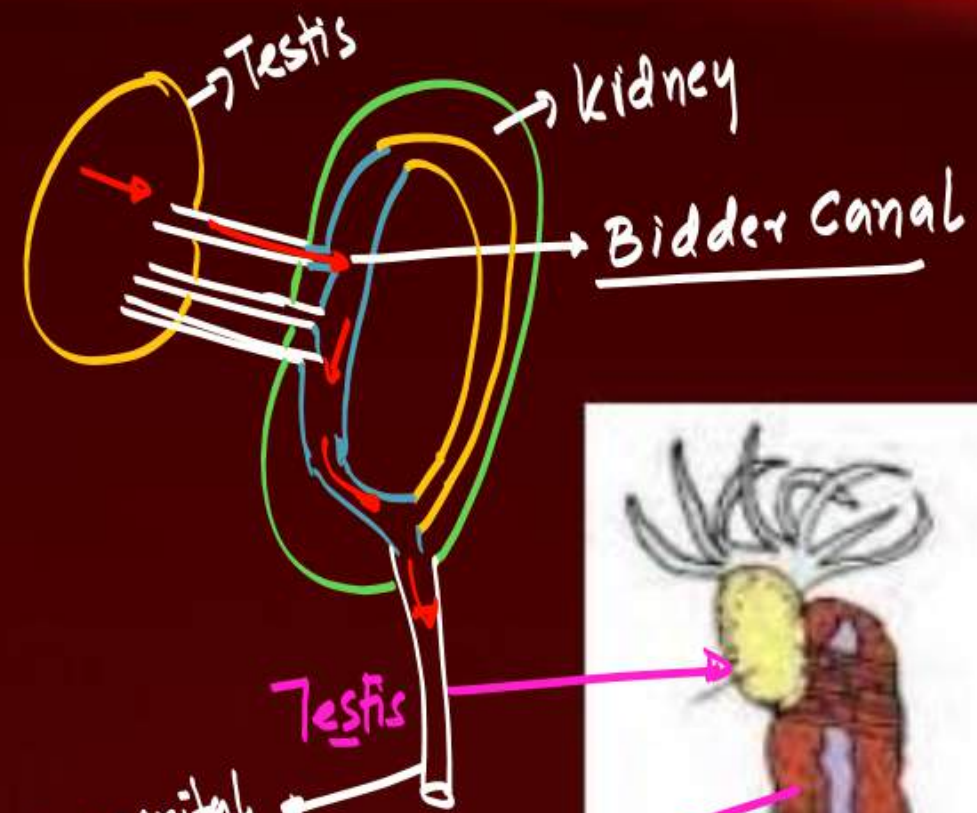
- Larval Stage = Tadpole → gills
 - Aquatic Environment → Ammonotelic
 (Ammonia)

Larva → Young Frog
 (Tadpole) ↴

Process - Metamorphosis

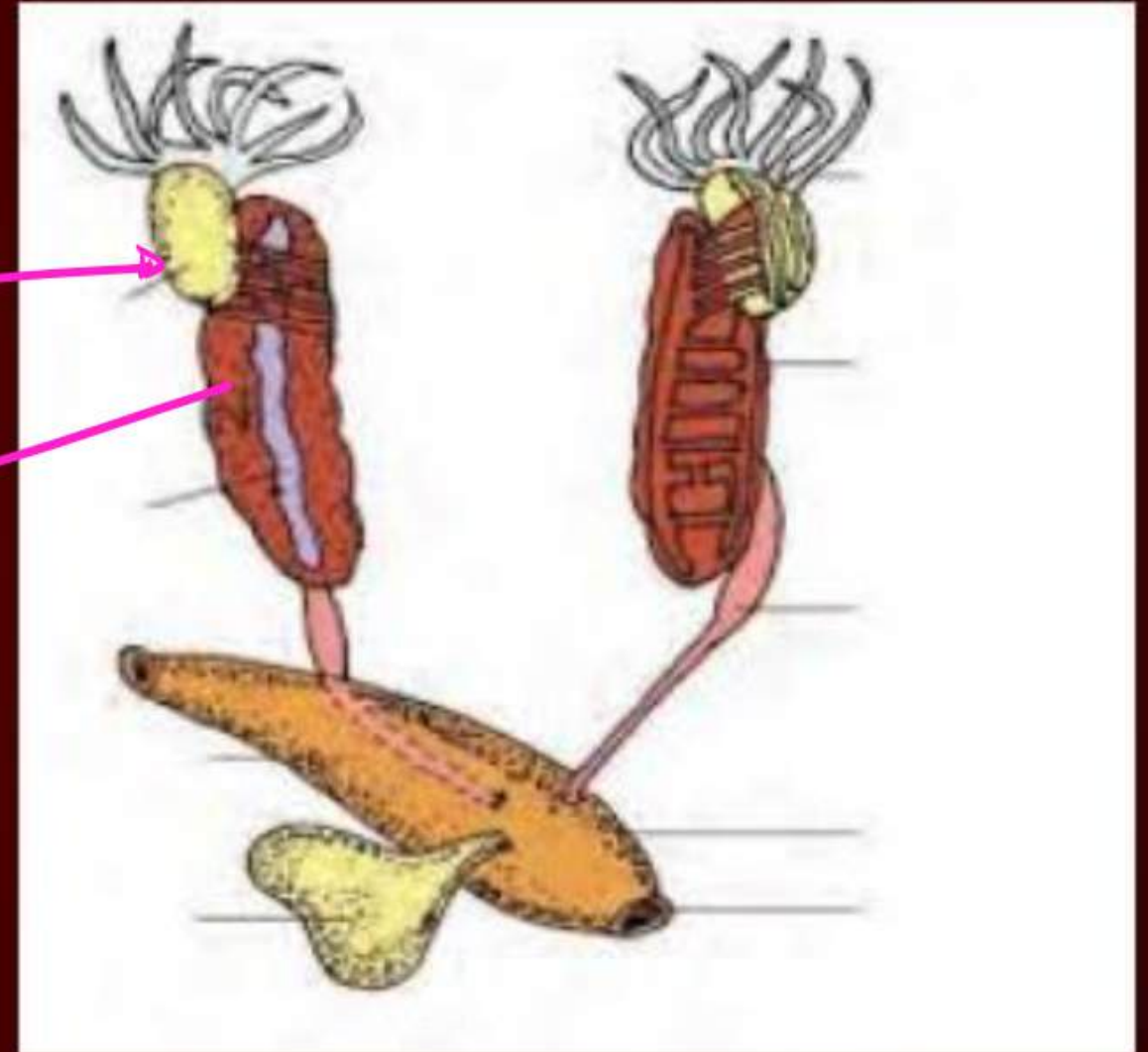
Male Reproductive System

- Testes (2 Testis) - Yellowish, Ovoid
 - Attached to kidney By (Mesorchium).



- Vasa efferentia (10-12)
 - Take Sperms from testis into Bidder's Canal.
- Bidder's Canal - Inside kidney
 - Receives Sperms from V. efferentia.
 - Joins collecting Duct and opens into Urogenital Duct
- Urogenital Duct (Ureter).
 - Carries Sperm into cloaca.

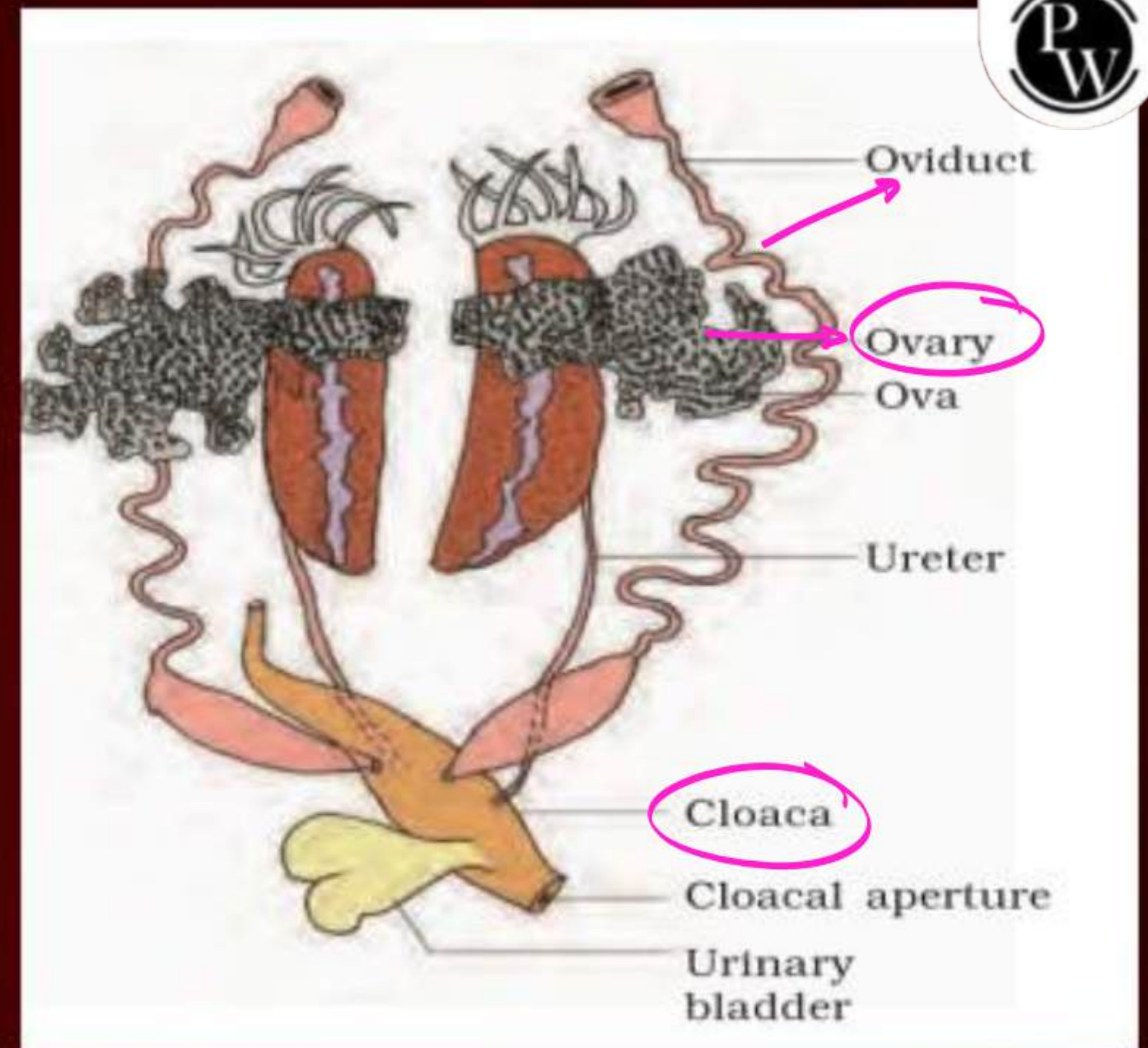
Urogenital Duct



→ Female Reproductive System

- Ovaries (2 ovary) - These are functionally not connected to kidneys.
 - Carry ovum directly in Cloaca.
- Oviducts - These are paired.
 - open into Cloaca.
 - Carry ovum from Ovary to Cloaca.
- Cloaca - Common chamber which also rectifies ovum.

(Female lays - 2500-3000 Eggs/Ova)



Biomolecules

Definition - These are Carbon containing organic Molecules present in living organism.

e.g CO_2 X - Not organic

(Glucose) - $C_6H_{12}O_6$ → Biomolecule [Carbon ✓, organic ✓, living organism ✓]

- (e.g) →
- Carbohydrates
 - Proteins
 - Lipids
 - Nucleic Acid
 - Enzymes (Mostly Proteins).

- Introduction
- Types (20-25 Min)
- Carbs "
- Proteins "
- Lipids "
- Nucleic "
- Enzyme "

Component	% of the total cellular mass
✓ Water	70-90
✓ Proteins	10-15
✓ Carbohydrates	3
✓ Lipids	2
✓ Nucleic acids	5-7
✓ Ions	1

→ Na^+ , Cl^- , Ca^{++} , Mg^{++} etc.

Element	✓ % Weight of Earth's crust Human body	
Hydrogen (H)	0.14	0.5
Carbon (C)	0.03	18.5 ✓ C
Oxygen (O)	46.6	65.0 O
Nitrogen (N)	very little	3.3 N
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

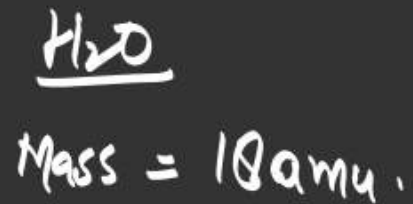
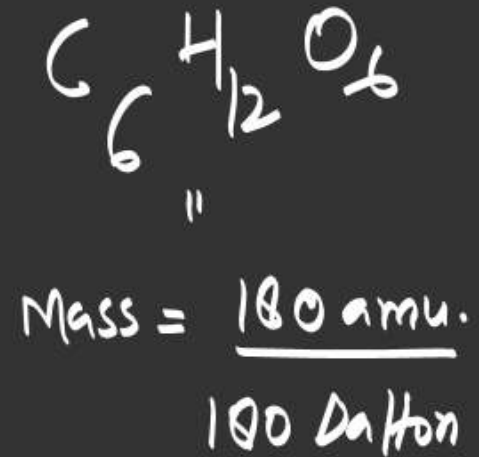
* Adapted from CNR Rao, *Understanding Chemistry*, Universities Press, Hyderabad.



Biomolecules

Classification
- Chemical Analysis.

$Q_N = \text{Protein}$ $\frac{1M}{x} \times 2M$



BiomicroMolecules
klas (Acid Soluble Pool)

- Size - $10 - 800 \text{ Dalton}$
 $\frac{1 \text{ Dalton} = 1 \text{ amu.}}{\text{Size}}$

eg. Monosaccharide (Glucose, Fructose)

- Nucleotides

- Amino Acids

Polymer

BioMacromolecules
klas (Acid Insoluble Pool)

Size = $> 10^6 \text{ Dalton}$

Lipids

Polysaccharide (Starch)

Nucleic Acid (DNA, RNA)

Protein

Ans- Protein is Made of Amino Acid.

- Macromolecule.

- $> 10^6 \text{ Dalton}$

- Acid Insoluble pool.



Living Tissue

Paste

Filter



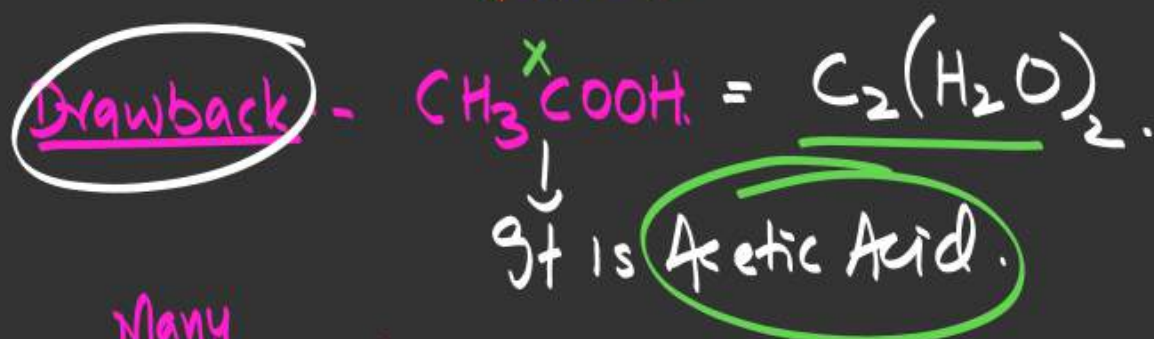
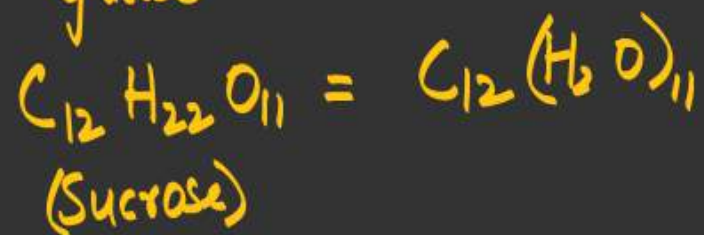
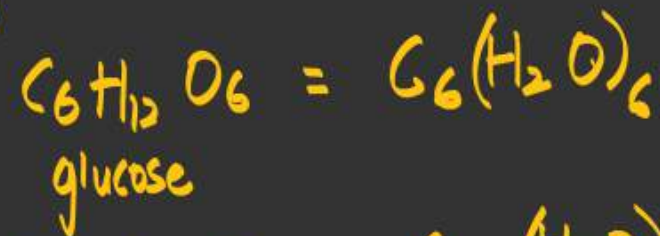
Retentate
(Acid Insoluble)

Filtrate
(Acid Soluble)



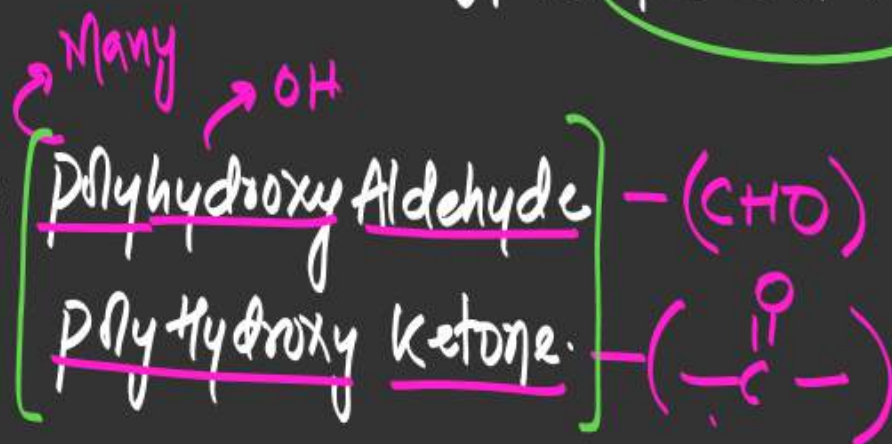
Carbohydrates $\frac{C}{H_2O}$ (literally - Hydrates of Carbon)

↳ Mainly - $C_n(H_2O)_m$ e.g.



New Definition =

These are



Q Diff Btw Glucose & Fructose.

Isomers

(Glucose)

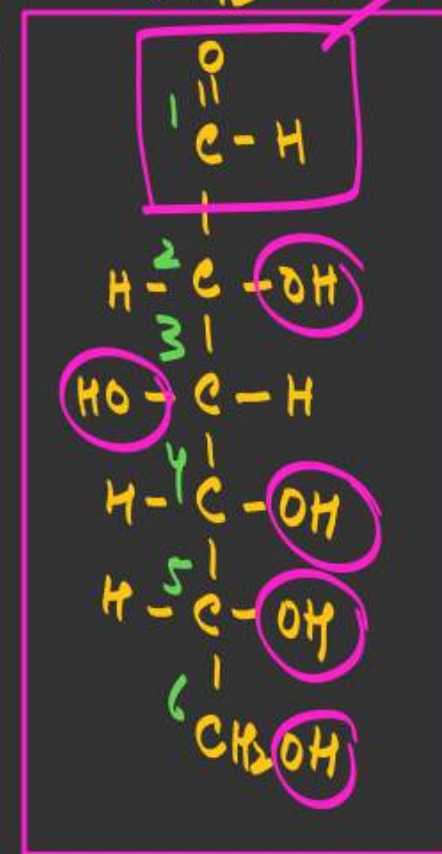


Aldehyde

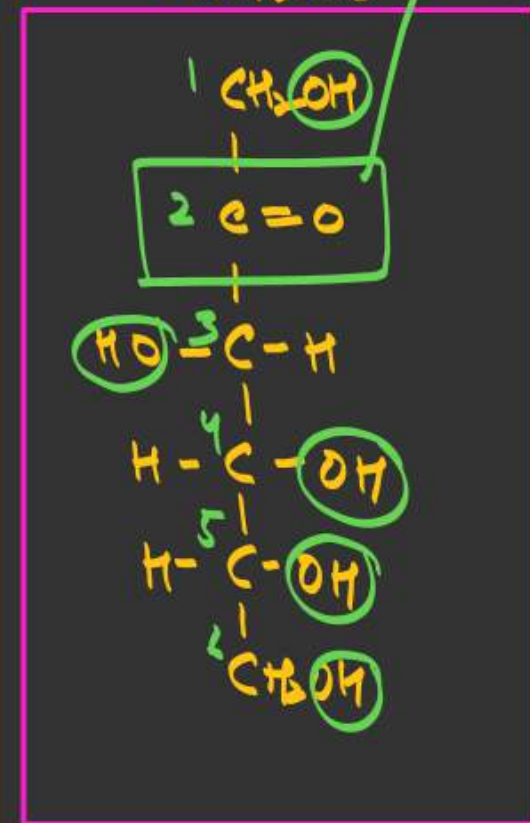
(Fructose)



ketone



Polyhydroxy
Aldehyde.



Polyhydroxy
ketone.



Classification of Carbohydrates

(On the Basis of Hydrolysis)

Bacterial
e.g. Pepsin, Trypsin

Monosaccharides **Reducing**

↳ Do not Hydrolyse.

↳ Simplest.

e.g. Triose (3C)

Tetrose (4C) $C_4H_8O_5$, $C_4H_8O_4$

✓ Pentose (5C) - Ribose, DeoxyRibose

✓ Hexose (6C) - Glucose, Fructose, Galactose

Heptose (7C)

Oligosaccharides

↳ They give 2-10 Units on Hydrolysis

↳ common - Disaccharide (Reducing) (Two Sugar) Except Sucrose

e.g. Lactose = Disaccharide of Glucose & Galactose
($C_{12}H_{22}O_{11}$)

= It is Milk Sugar

= Bond - Glycosidic (1-4) Bond

→ Sucrose = Disaccharide of Glucose & Fructose
($C_{12}H_{22}O_{11}$)

= It is Table Sugar, Cane Sugar

= Bond Glycosidic (1-2)

→ Maltose = Disaccharide of Glucose & Glucose
= It is Malt Sugar
= Bond - Glycosidic

(Non-Reducing) Polysaccharides

↳ They give > 10 Units

They can be

↳ Homopolysaccharides (Made of only one type of sugar)

e.g. Starch - Plants (stored) - Glucose

Cellulose - plant (cell wall) - (Glucose) - Structural

Glycogen - Animals, stored, Glucose

Chitin - Fungal cell wall, Exoskeleton of Insects (N-acetyl glucosamine)

Inulin
↳ Stored (Fructose).
e.g. Dahlia Plant

↳ Heteropolysaccharides (Made of more than one type of sugar)

Proteins

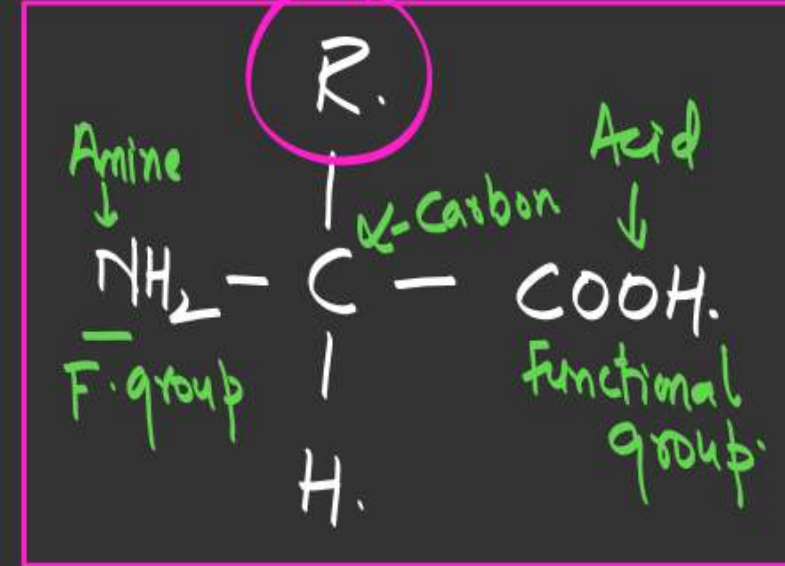
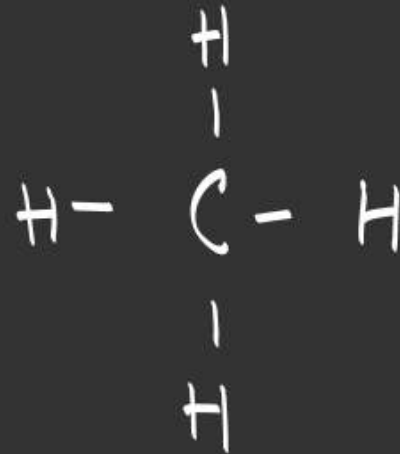
→ Heteropolymer of α -Amino Acid

↓ Made of more than one type

Because Amine & Acid are Attached to α -Carbon

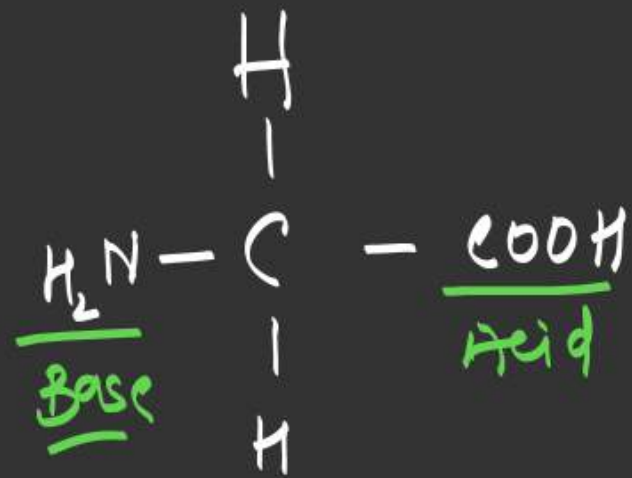
- Amino Acid - It is Substituted Methane

→ Every A.A Acid Different R group

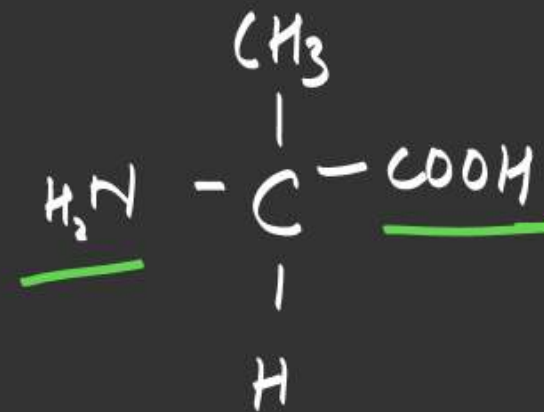


Basic Structure of Amino Acid

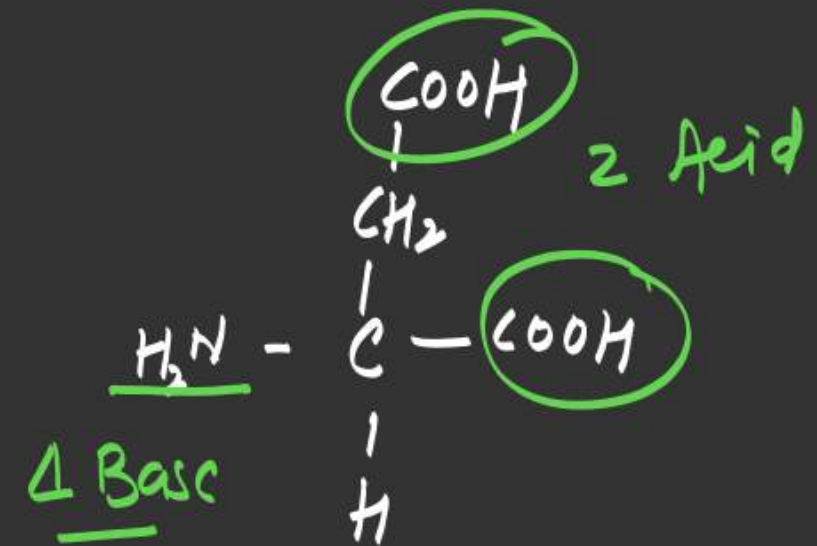
- There are 20 Amino Acids present in Proteins.



→ Glycine
(simplest)



→ Alanine



→ Aspartic Acid.

Classification of Amino Acid → (R-group Basis)

→ Nutritional Value

Essential

- Can't be synthesized in Body
- from Diet

✓ → Lysine, Threonine.

✓ → Leucine

✓ → Isoleucine

→ Valine

- Arginine

- Histidine

- Methionine

- Tryptophan

- Phenylalanine

Non-Essential

- Can be synthesized in Body

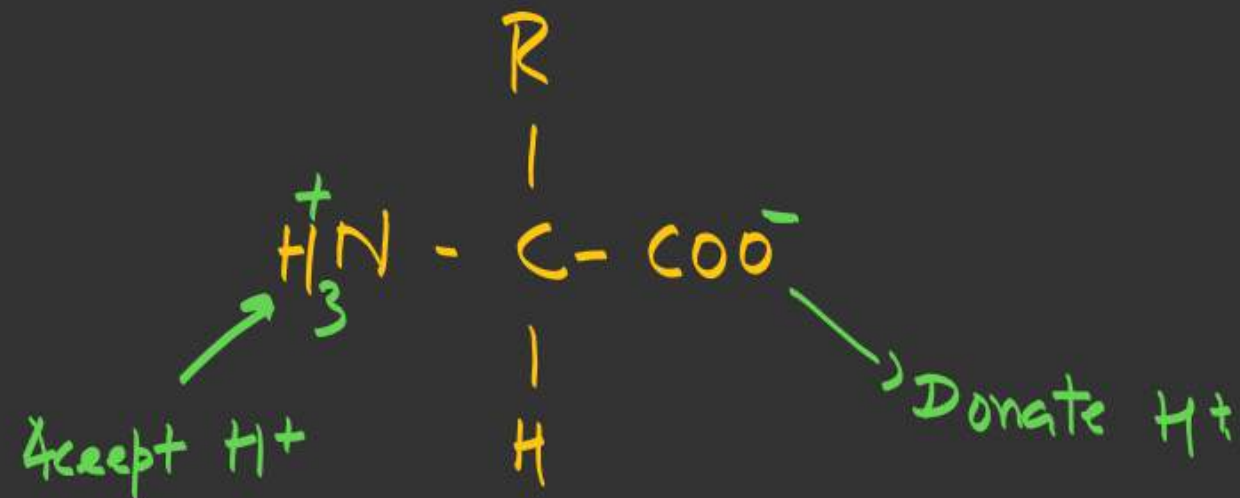
- Glycine.

- Glutamic Acid

- Aspartic Acid

- ✓ ① Neutral - Glycine, Alanine,
- ✓ ② Acidic - Aspartic Acid, Glutamic Acid,
- ✓ ③ Basic - Lysine, Arginine
- ④ OH containing - Serine, Threonine
- ⑤ Sulphur containing - Cysteine, Methionine,
- ⑥ Aromatic (Ring) - Tyrosine, Phenylalanine.

→ Amino Acid Form Zwitter Ion
 (Having Both $\oplus$ & $\ominus$ charge)
 - It is Electrically Neutral



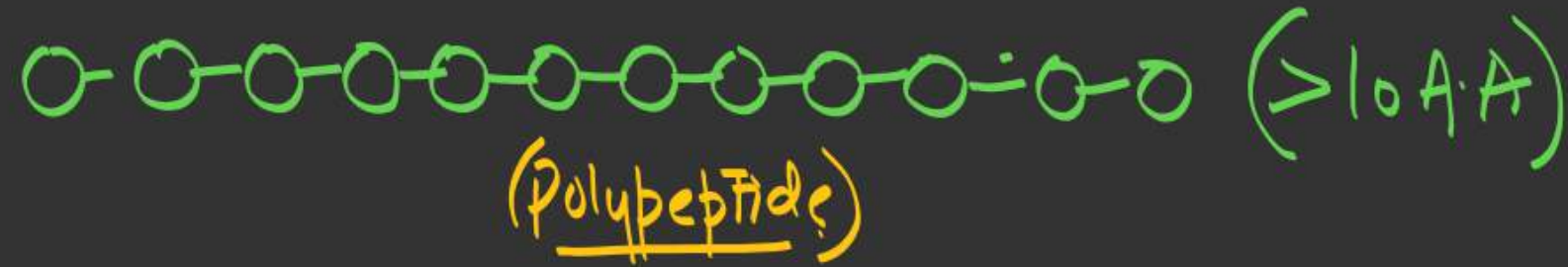
pH = Isoelectric pH

Protein - It is a long chain polymer of Amino Acids. (Polypeptide)

e.g



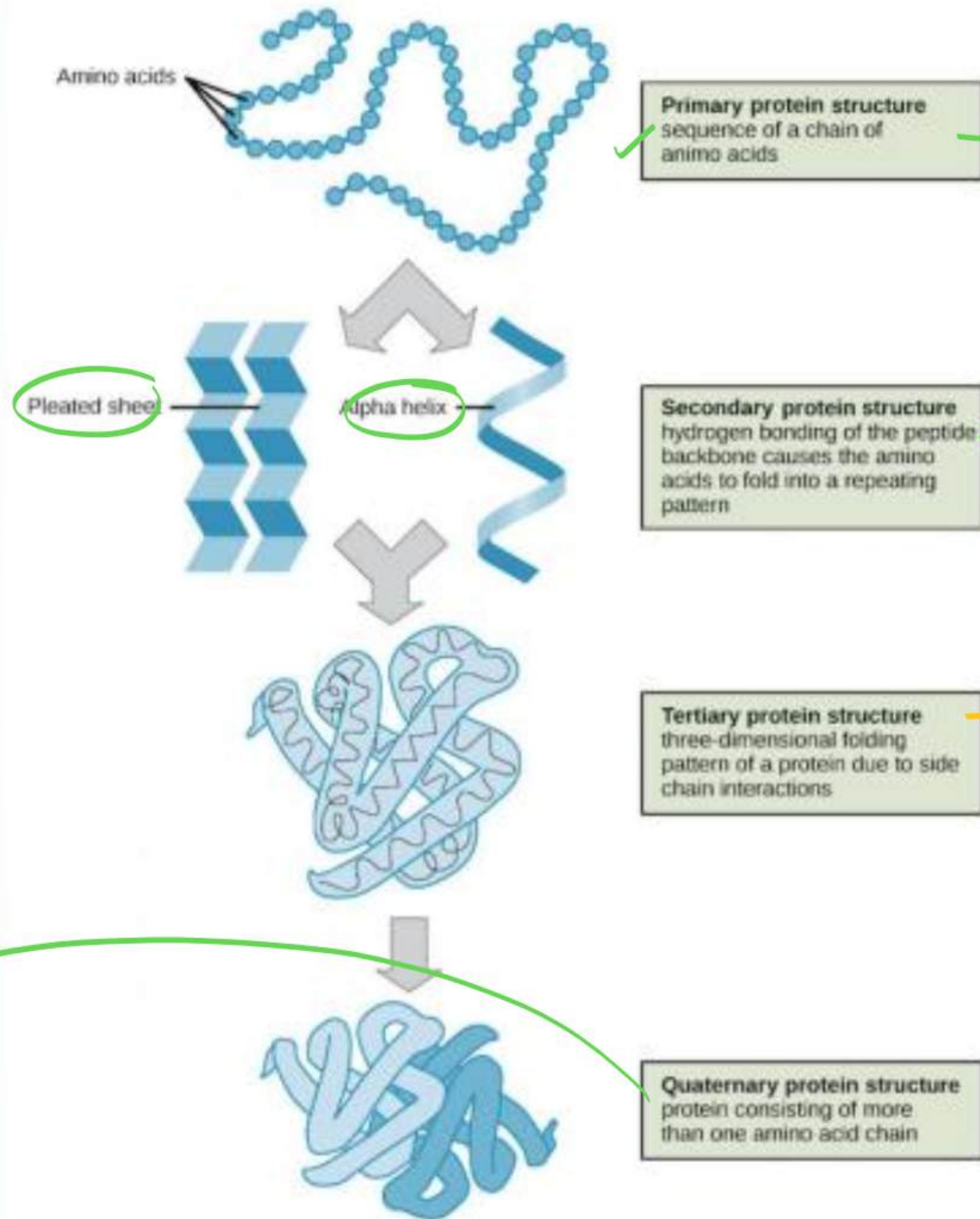
Bond peptide Bond



Carbohydrate Bond = ?
glycosidic

AMINO ACID			
Nonpolar, aliphatic R groups			
	Glycine	Alanine	Valine
	Leucine	Methionine	Isoleucine
	Serine	Threonine	Cysteine
Polar, uncharged R groups			
	Proline	Asparagine	Glutamine
AMINO ACID			
Positively charged R groups			
	Lysine	Arginine	Histidine
Negatively charged R groups			
	Aspartate	Glutamate	
Nonpolar, aromatic R groups			
	Phenylalanine	Tyrosine	Tryptophan

H/W - Fibrous →
Globular →



→ Primary

- Linear chain
- Biologically- Non Function

→ Collagen, Fibrin (Silk)

→ Secondary structure

- α - Helix
- β - Pleated Sheets
- Hydrogen Bonding



→ Tertiary Structure (Myoglobin)

- Overall 3D coiling on its



- Bonds - Ionic
- Disulphide
- Hydrogen etc

$\alpha_2 \beta_2$
Quaternary (Haemoglobin)

- More than one tertiary Units.

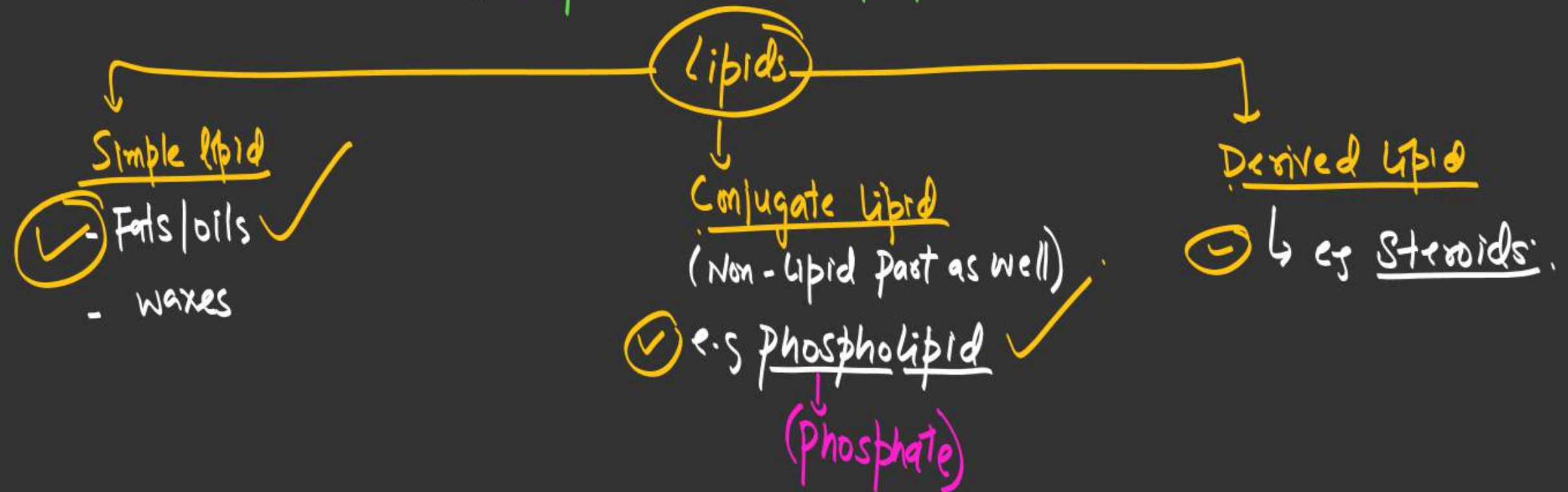
Lipids

↳ These are Hydrophobic (Hate Water)
Insoluble in H₂O

↳ Are Acid Insoluble pool.

↳ But Not (Strictly) Biomacromolecules (Size is smaller)

↳ Lipids are not polymeric.



→ solid at Room temp.
 - Fats/oils -
 ↓
 Liquid at Room temp.

→ These are triglycerides in Nature.

→ Triglyceride is Made of 3 Fatty Acid + 1 glycerol.

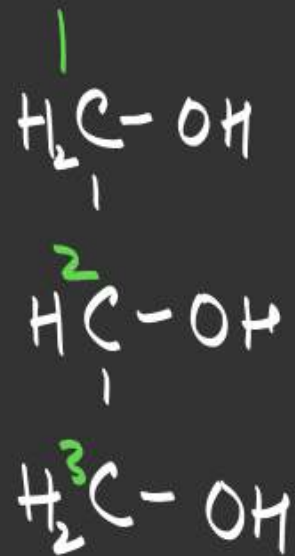
→ Fatty Acid is Long chain Carboxylic Acid.

e.g. $\text{CH}_3 - (\text{CH}_2)_{14} - \text{COOH}$. (16C)

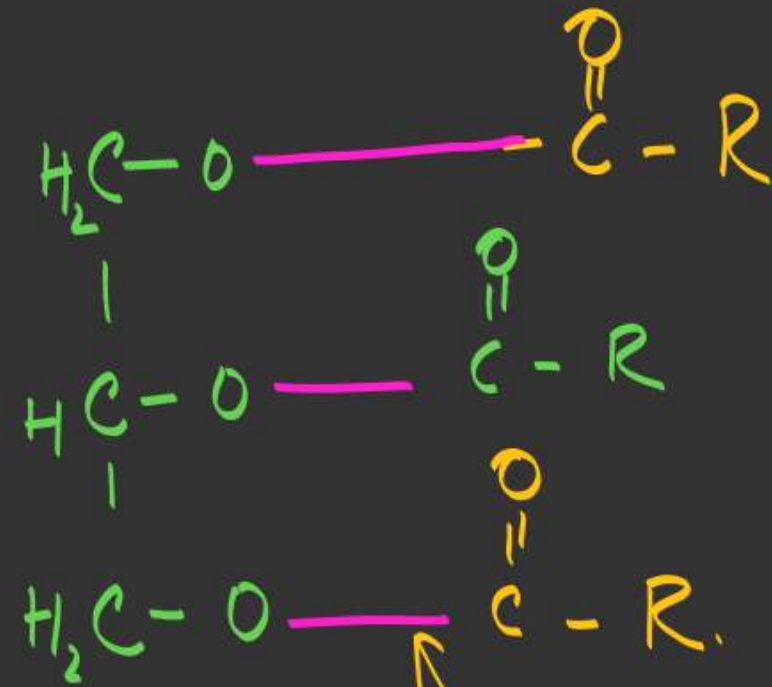
- palmitic Acid.

e.g. - Stearic Acid (18C).

→ Glycerol - Propan 1,2,3 triol



Structure of Triglyceride



phosphatid

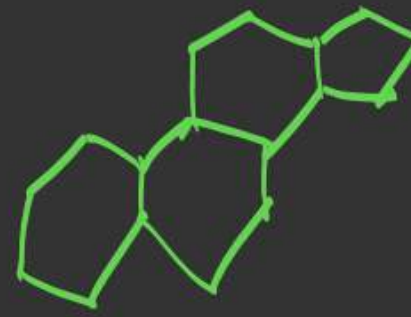
Phospholipid - It is made when 2 Fatty Acids and one phosphate groups
Attaches to Glycerol



- present in cell Membrane
- e.g Lecithin, Cephalin.

Steroid - These are derived lipids

- They contain Steroid Nucleus



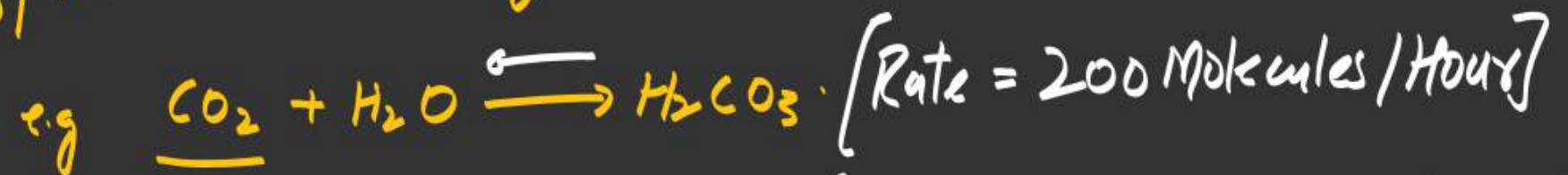
e.g. cholesterol (Most Abundant in Animals)

- Vit D.

Hormone. $\left\{ \begin{array}{l} \cdot \text{Sex Hormones} \cdot \\ \cdot \text{Cortisol} \\ \cdot \text{Aldosterone} \end{array} \right.$

Enzymes

- Alters Rate of Reaction.
- ↳ These are Bio catalyst
- Enzymes increase Rate of Biological Reactions.



(Carbonic Anhydrase) Enzyme → [Rate = 6 Lakh Molecules/sec].

Rate Increased 10 Million Times

Nature of Enzyme

- Mostly Made of Proteins (tertiary Structure).
- Some/Few are Made of RNA (e.g. Ribozymes/Ribonuclease)

- Enzymes lower down Activation Energy of Reaction

↳ Minimum Energy Required by Enzymes, so that they can form Products

→ Factors Affecting Enzyme Activity

① Substrate Concentration

② Temp.

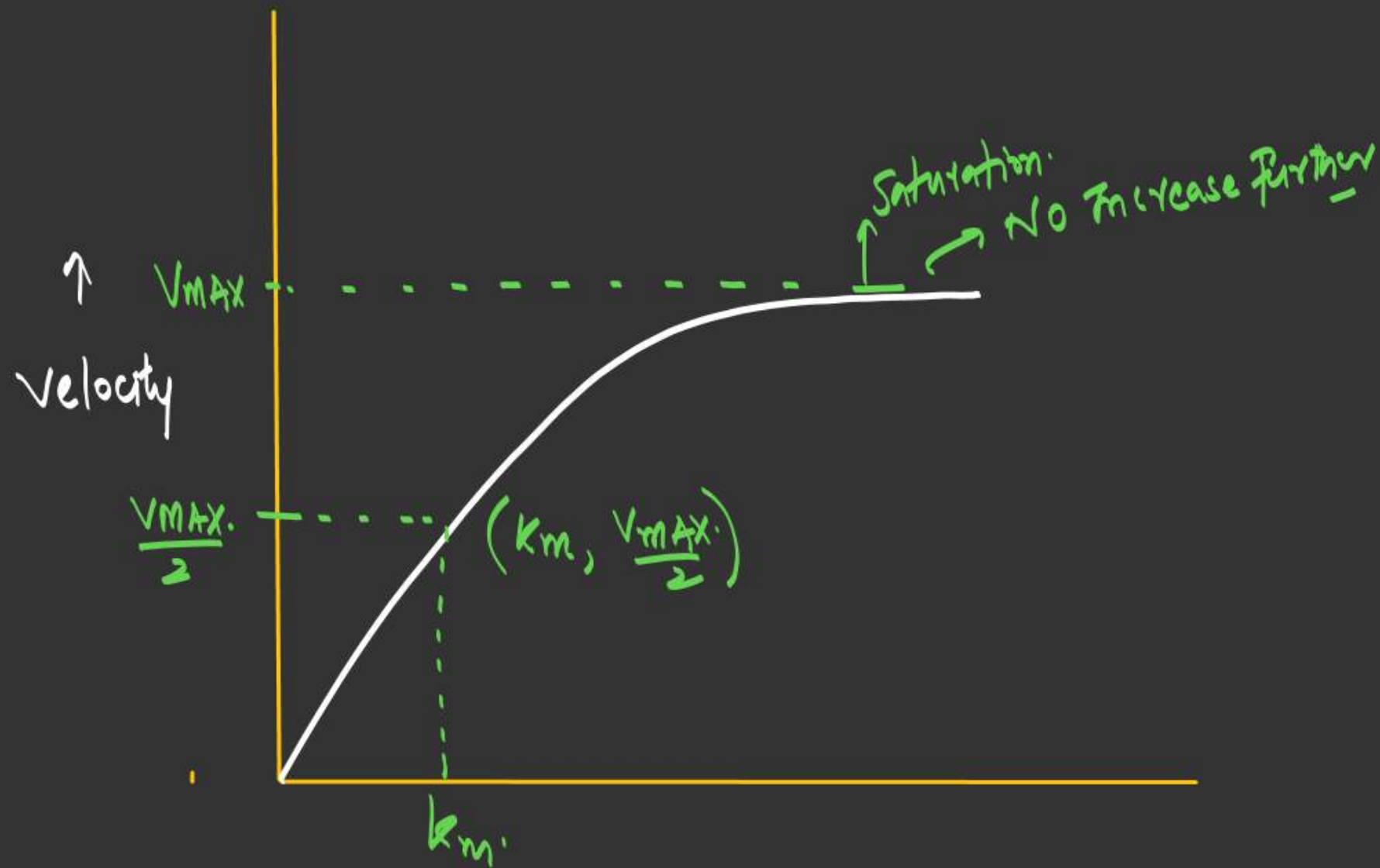
③ pH.

④ Inhibitor

① Substrate Concentration

(1000 E = 1000 Active Sites)

→ Here Speed of Reaction Increasing on increasing Substrate Conc. to a point of Saturation.

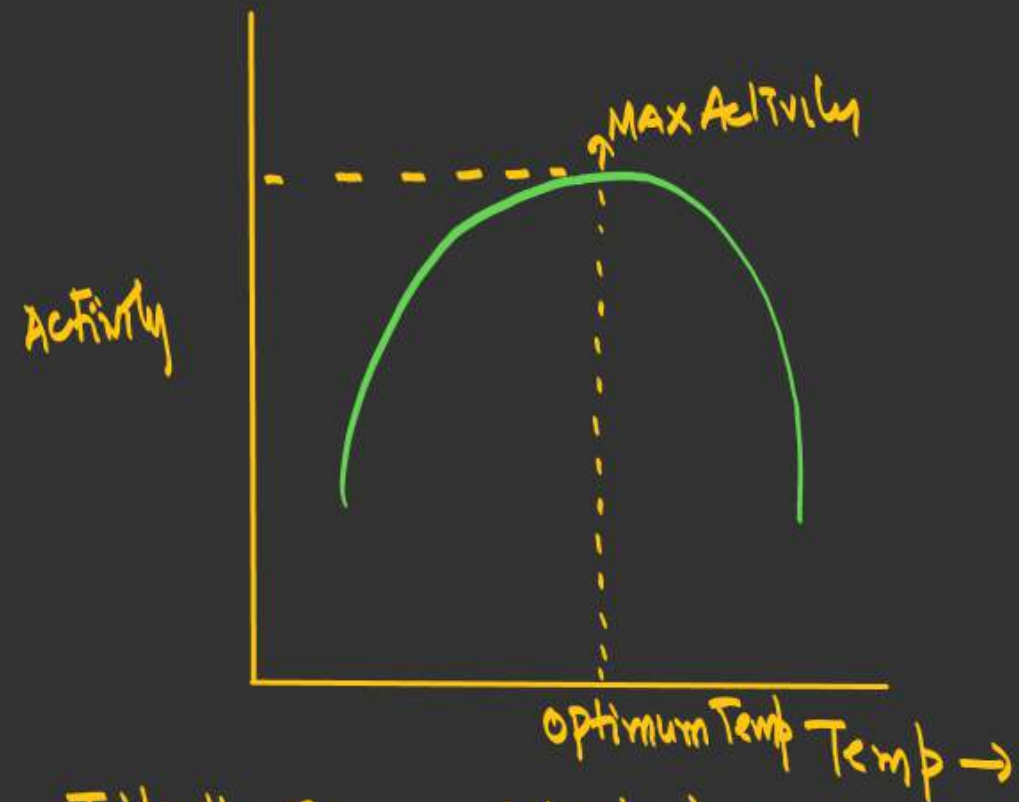


Substrate Conc.

$K_m = \text{Micheal Menten Const}$

① Temp

→ Bell Shaped Curve

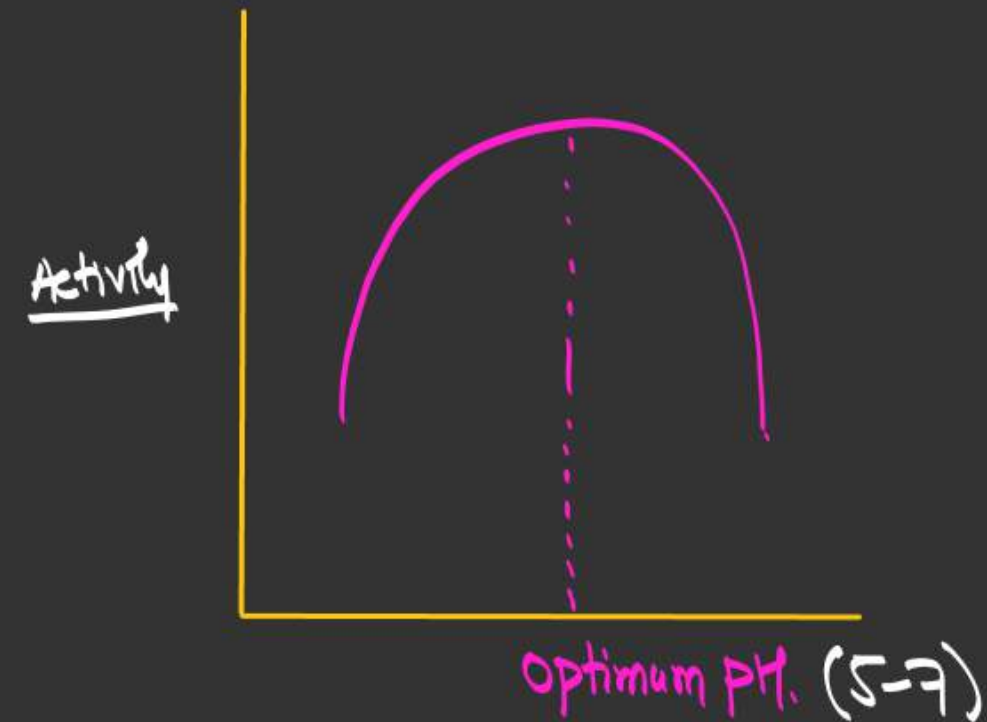


Initially Enzyme Activity increases on increasing temp.

- Then it decreases Due to Denaturation/Damage to Enzyme

③ pH

Same Shaped Curve



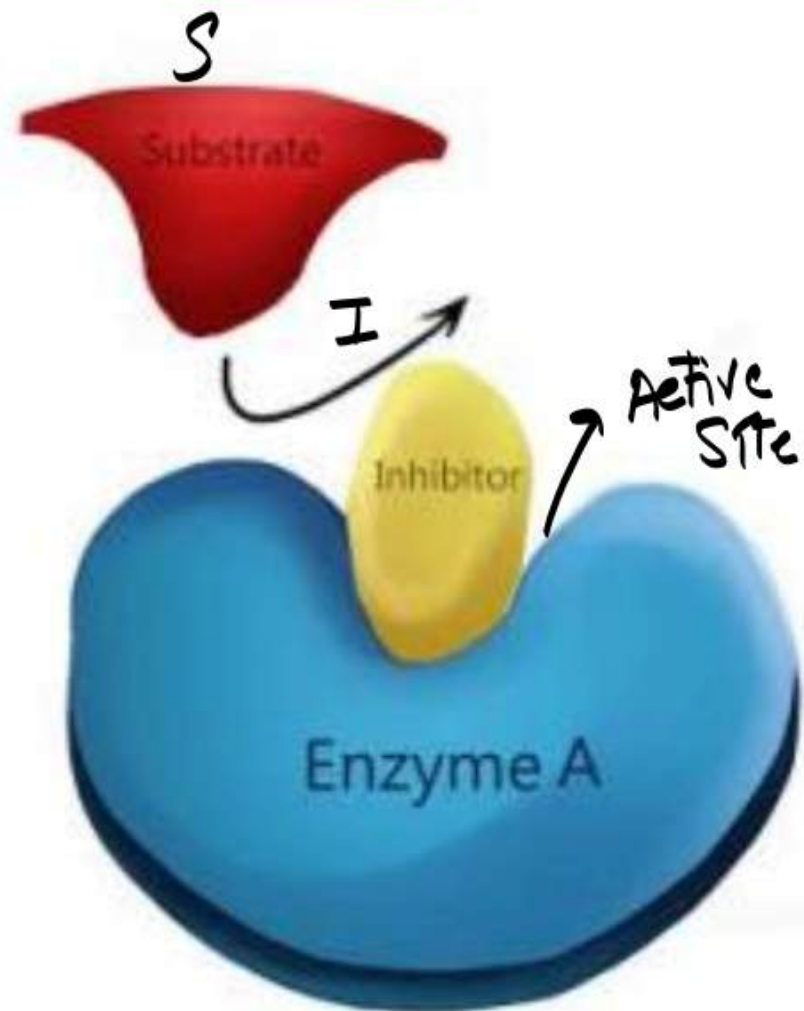
→ Low Activity at Extreme Acid & Base
It damages Enzyme

⑮ Inhibitor - Causes Inhibition
when Enzyme Activity is Reduced or Blocked.



Inhibition

A) Competitive Inhibition



B) Non-competitive Inhibition



Here $[S]$ & $[I]$ are Structurally Similar

→ I Binds to Active Site

→ Blocks Enzyme Activity

eg E = Succinic dehydrogenase

S = Succinic Acid

I = Malic Acid
Oxalic Acid

→ Here I & S are Structurally Different

- I Binds other than Active Site

- But Still Blocks Enzyme Activity

eg CN inhibit cytochrome oxidase:

Classification

Simple Enzymes

- only made of protein
- e.g. pepsin

Conjugate Enzymes

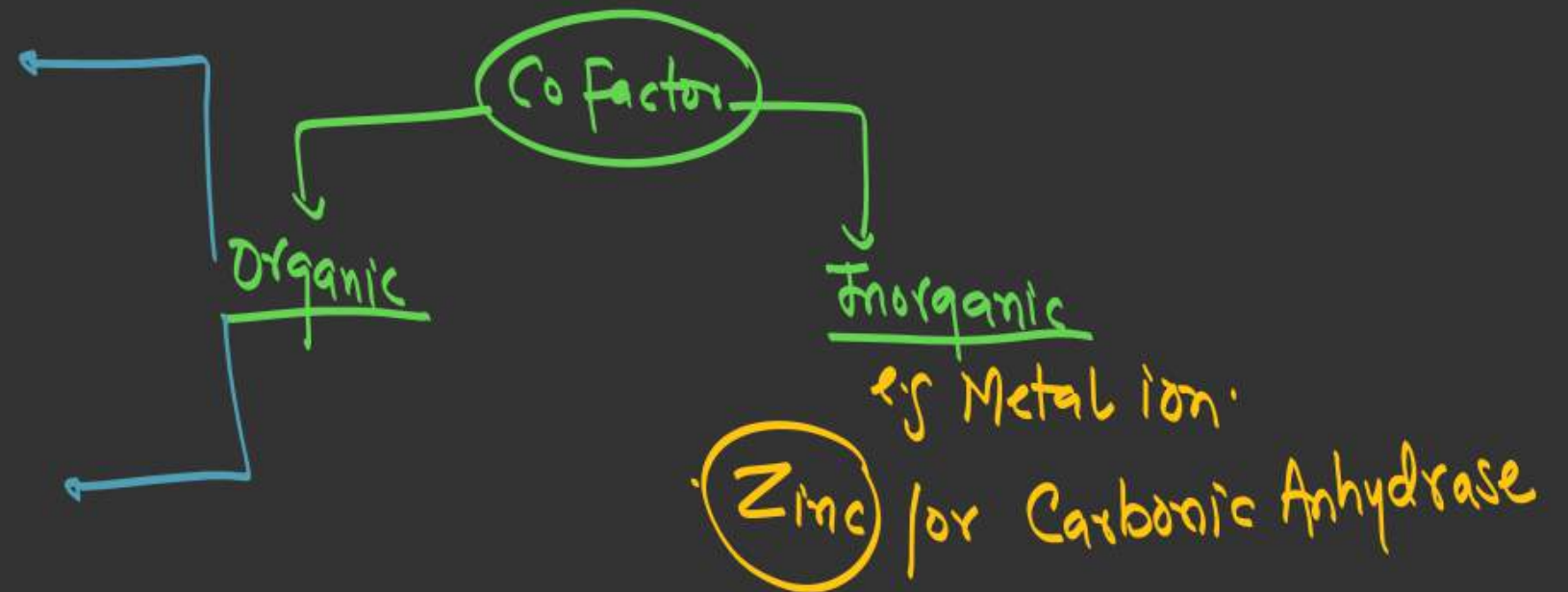
↳ made of non-protein = co factor

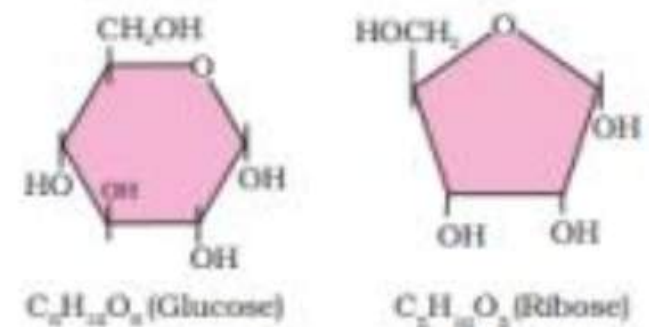
Two parts

protein + nonprotein = Enzyme
(Apoenzyme) + (co-factor) = Holoenzyme

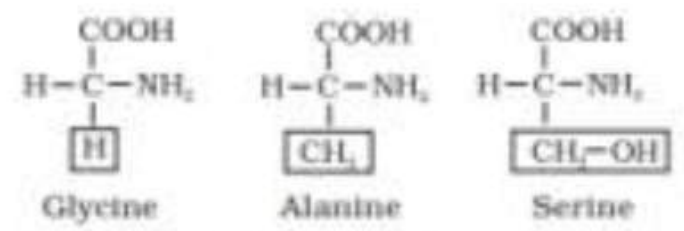
(e.g) Heme in peroxidase/catalase - Prosthetic group
- Tightly Attached
- Always

(e.g) NAD, NADP etc - Co-enzyme
- Loose Attached
- During Reaction

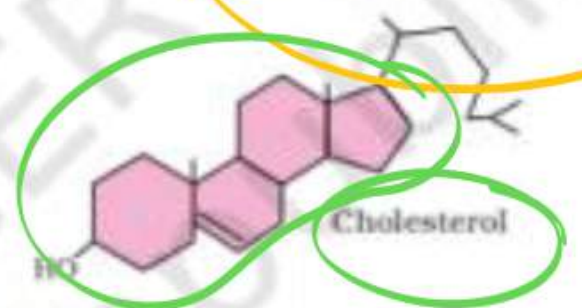
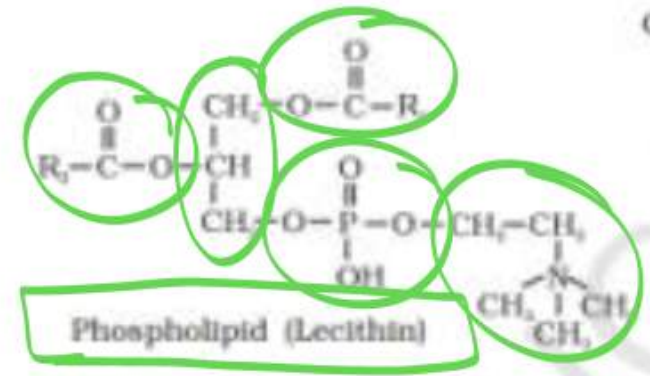
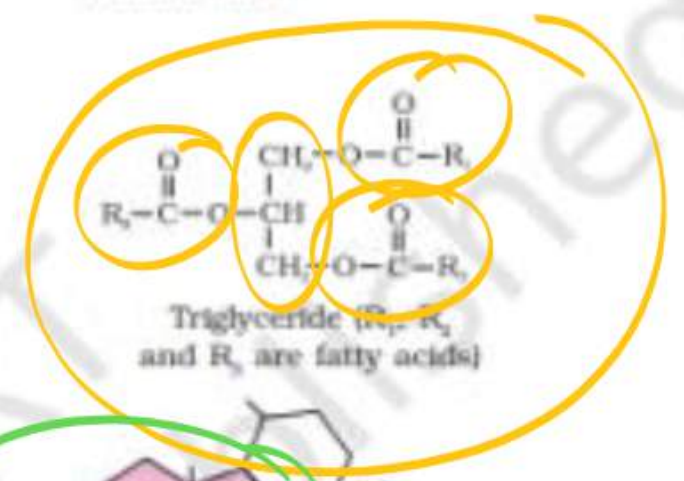
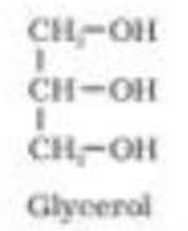
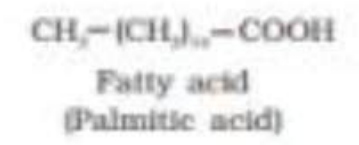




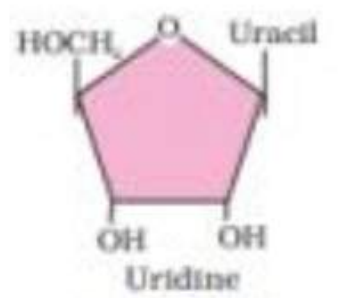
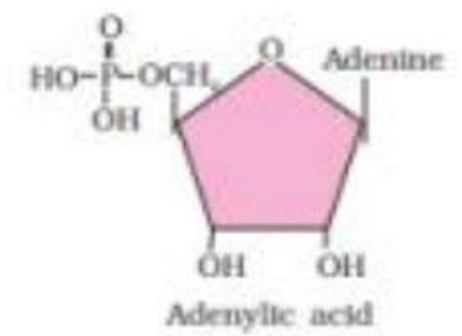
Sugars (Carbohydrates)



Amino acids



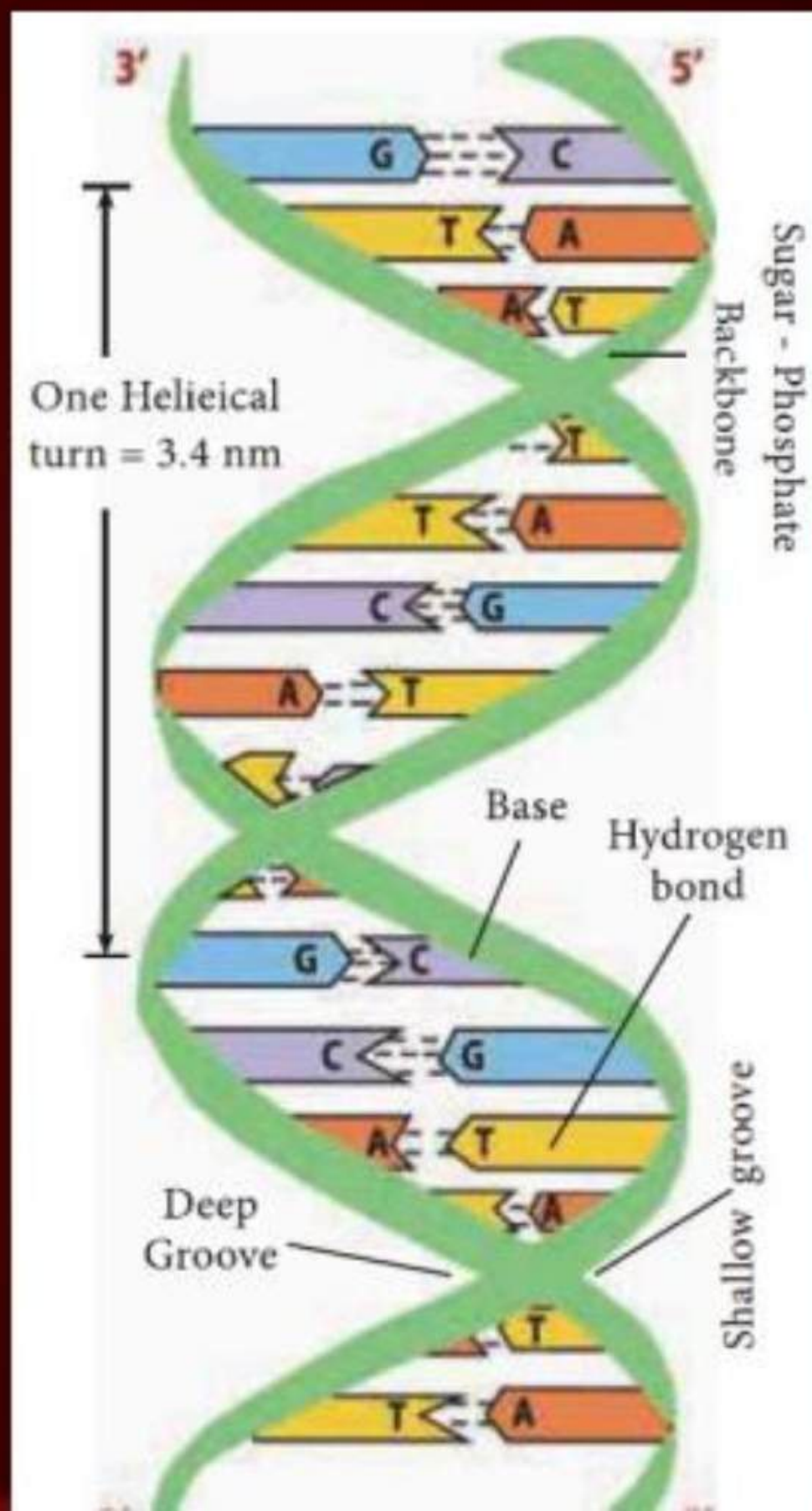
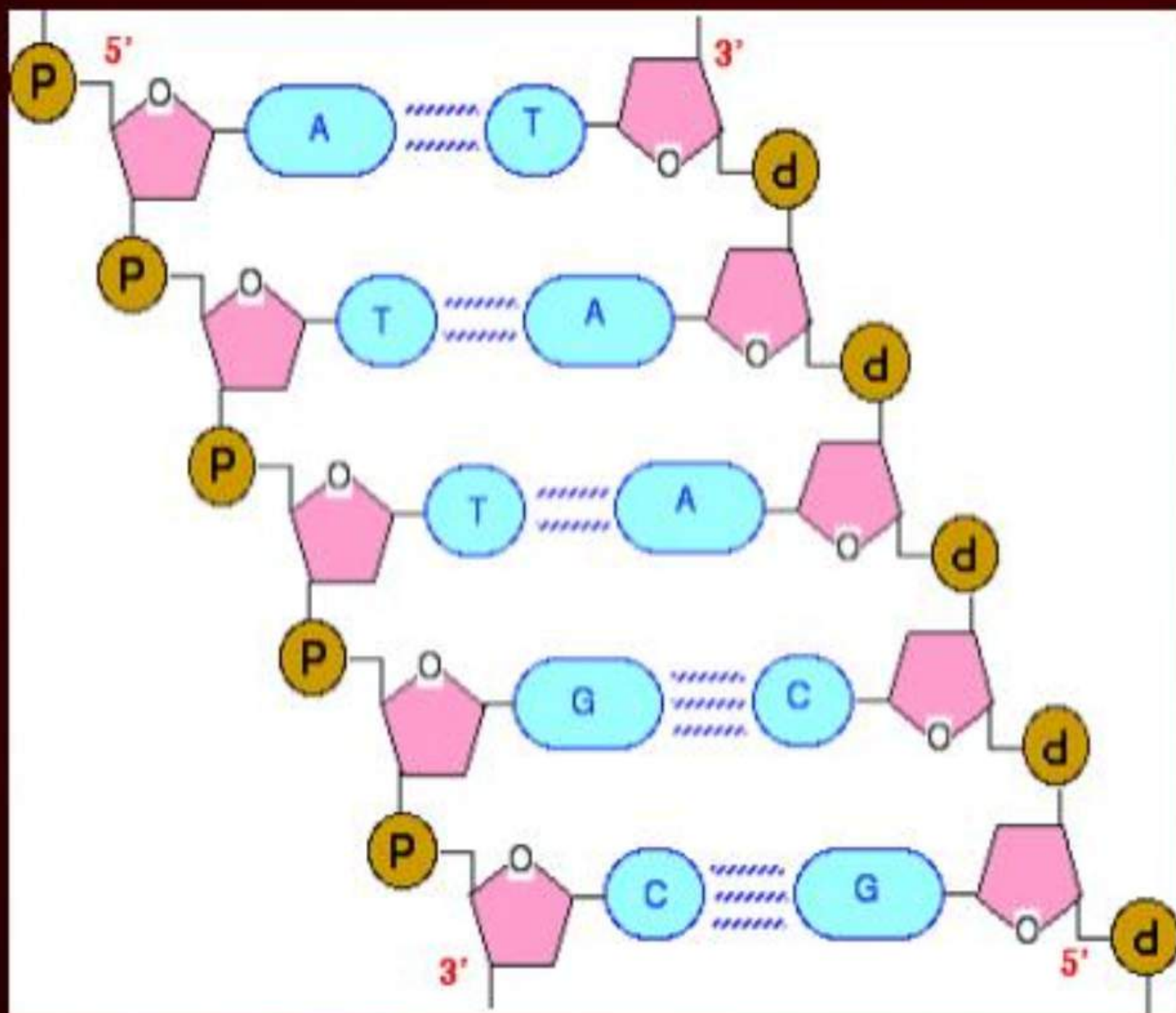
Fats and oils (lipids)



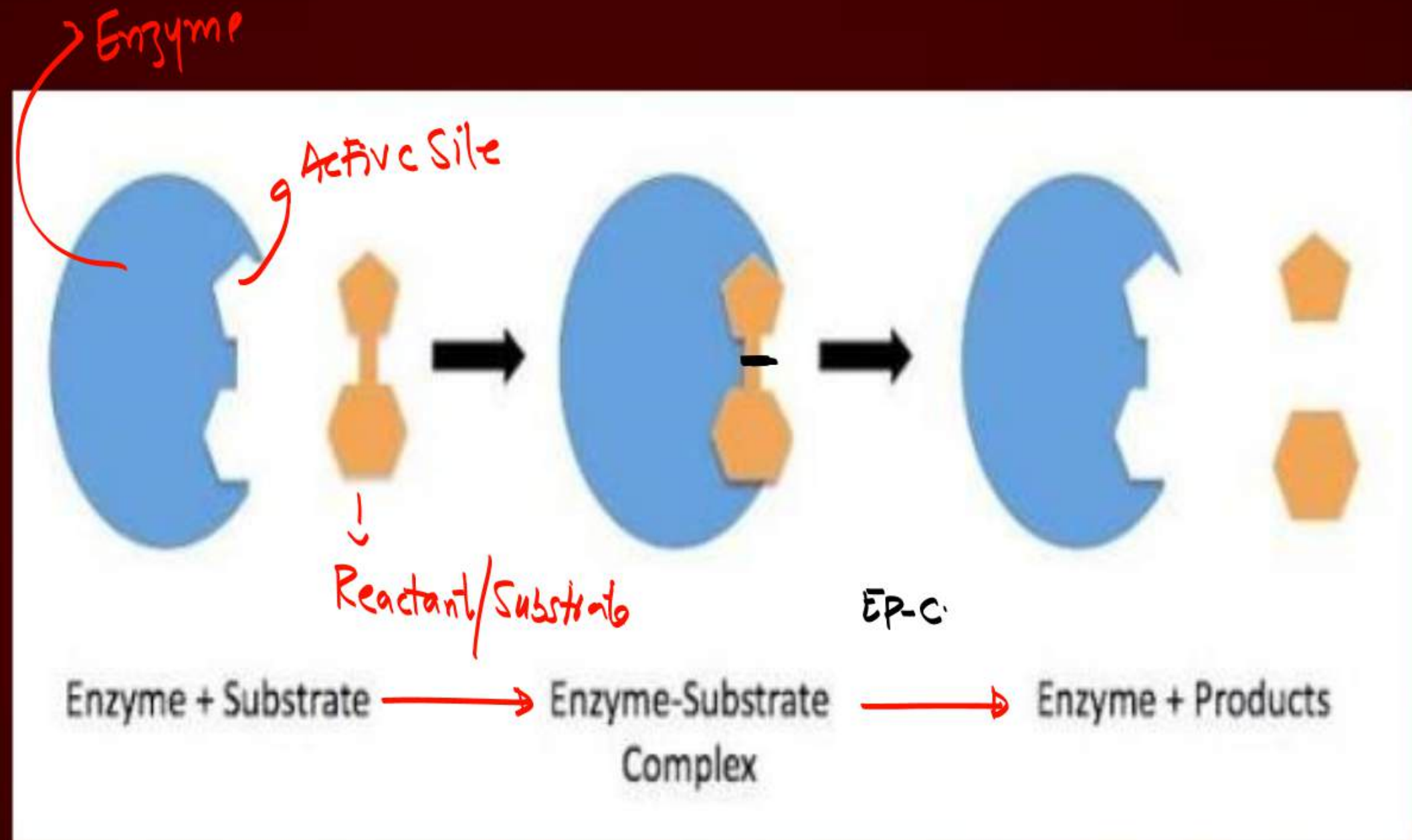
Nitrogen bases

Nucleosides

Nucleotide

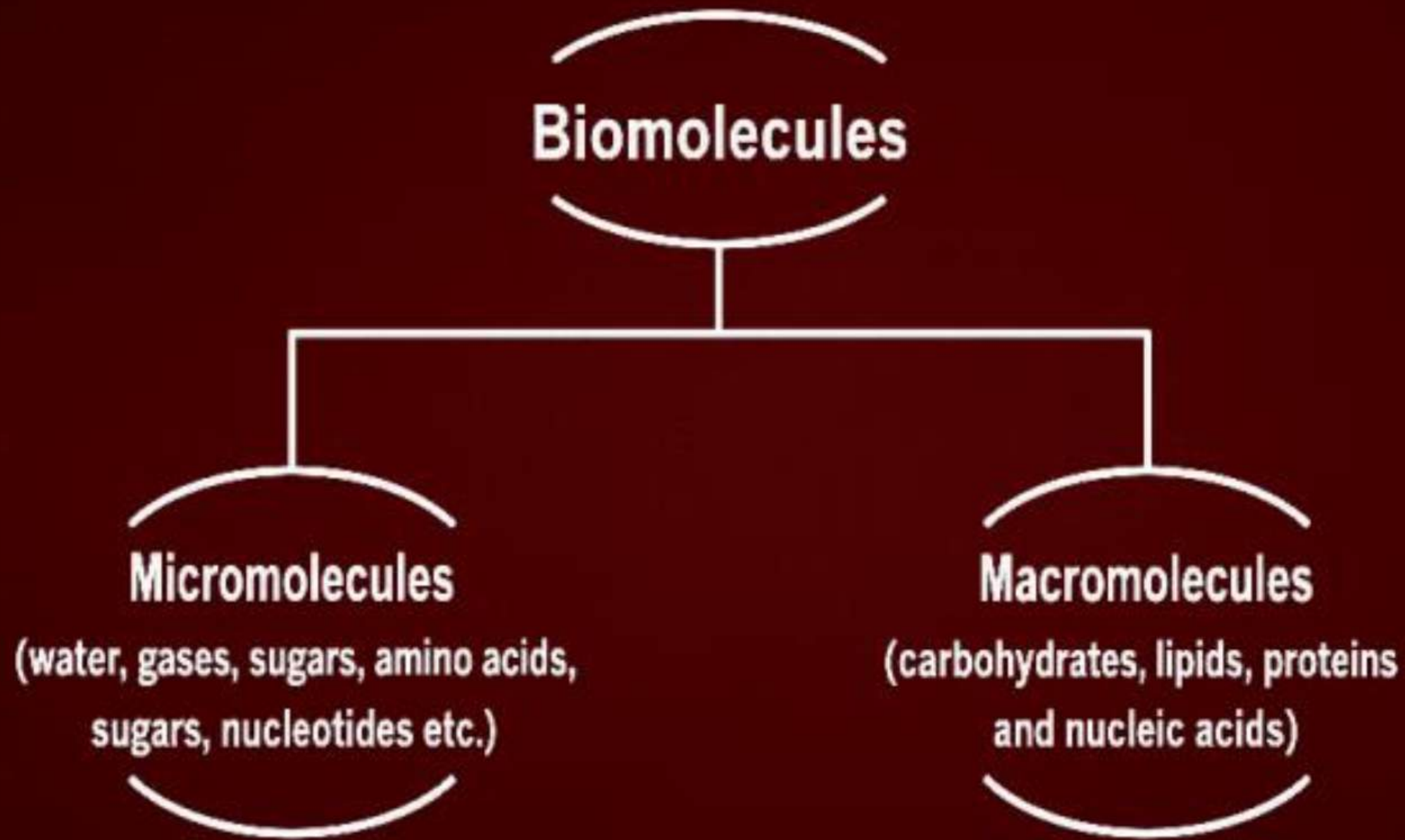


(Mechanism of Enzyme Action)





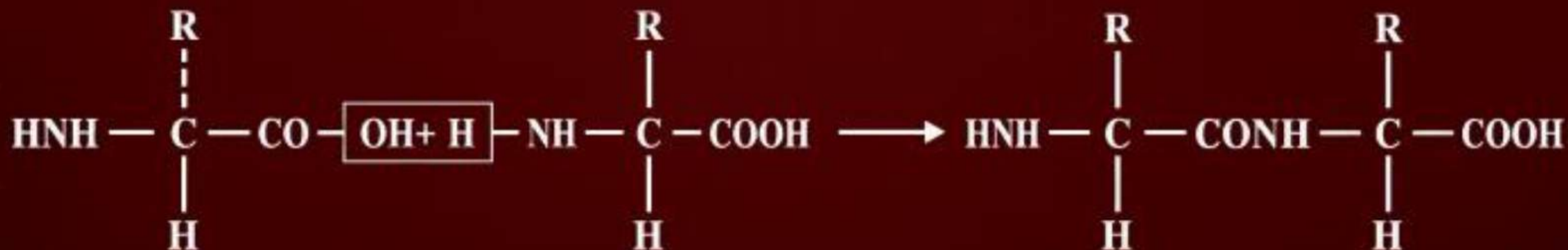
- ❑ Chemicals or molecules present in the living organism are known as biomolecules. Biomolecules are divided into two types- inorganic and organic.
- ❑ Inorganic biomolecules includes minerals, gases and water and organic biomolecules includes carbohydrates, fats, proteins, nucleic acids, vitamins, etc.
- ❑ Different biomolecules can be classified as aldehyde, ketones and aromatic compounds as chemical forms. The amino acids, nucleotides and fatty acids can be classified as biochemical forms.





Biomolecules – Micromolecules and Macromolecules

- * Except lipids, macromolecules are formed by polymerization of sub-units called monomers.
- * Proteins are polymers of amino acids. Amino acids are linked by **peptide bond** formed by dehydration between COOH group of one amino acids and NH_2 group of next with the removal of H_2O .

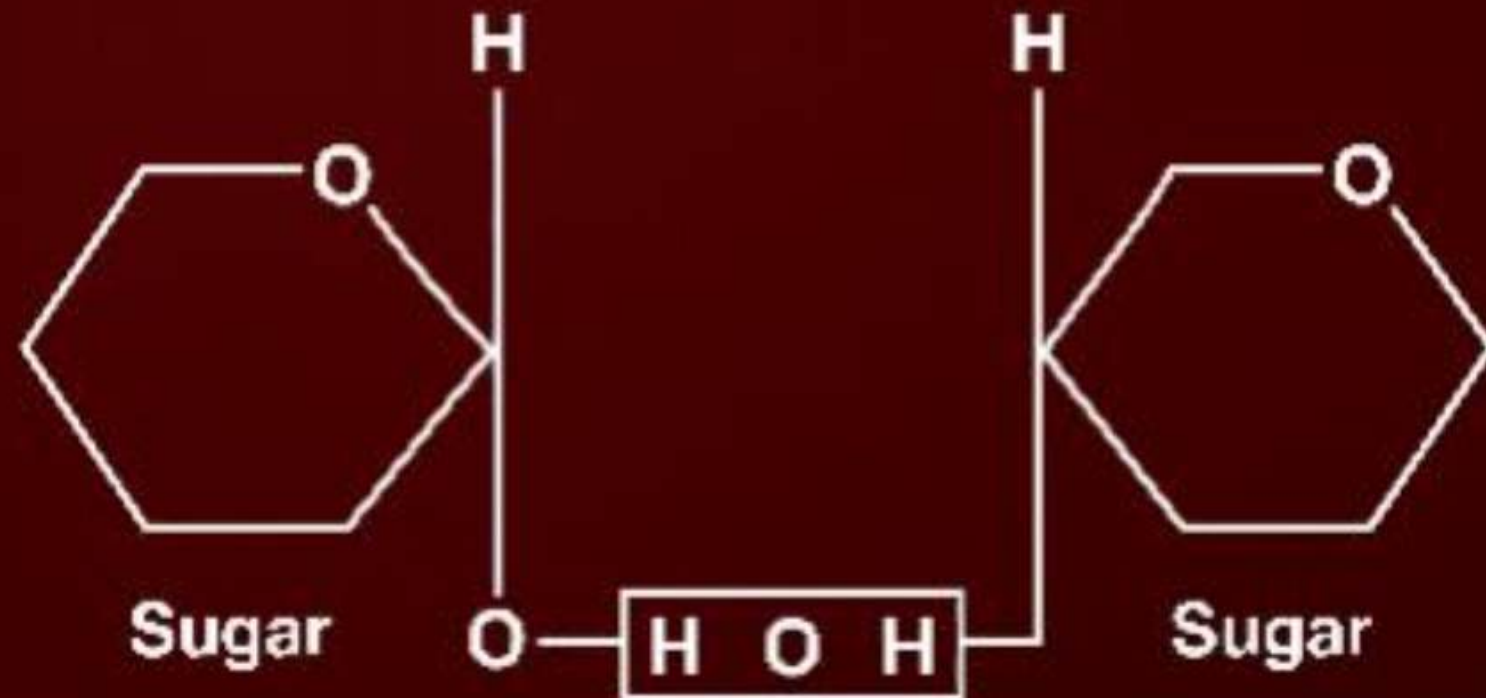




Nucleic Acids and Polysaccharides — Bond Formation



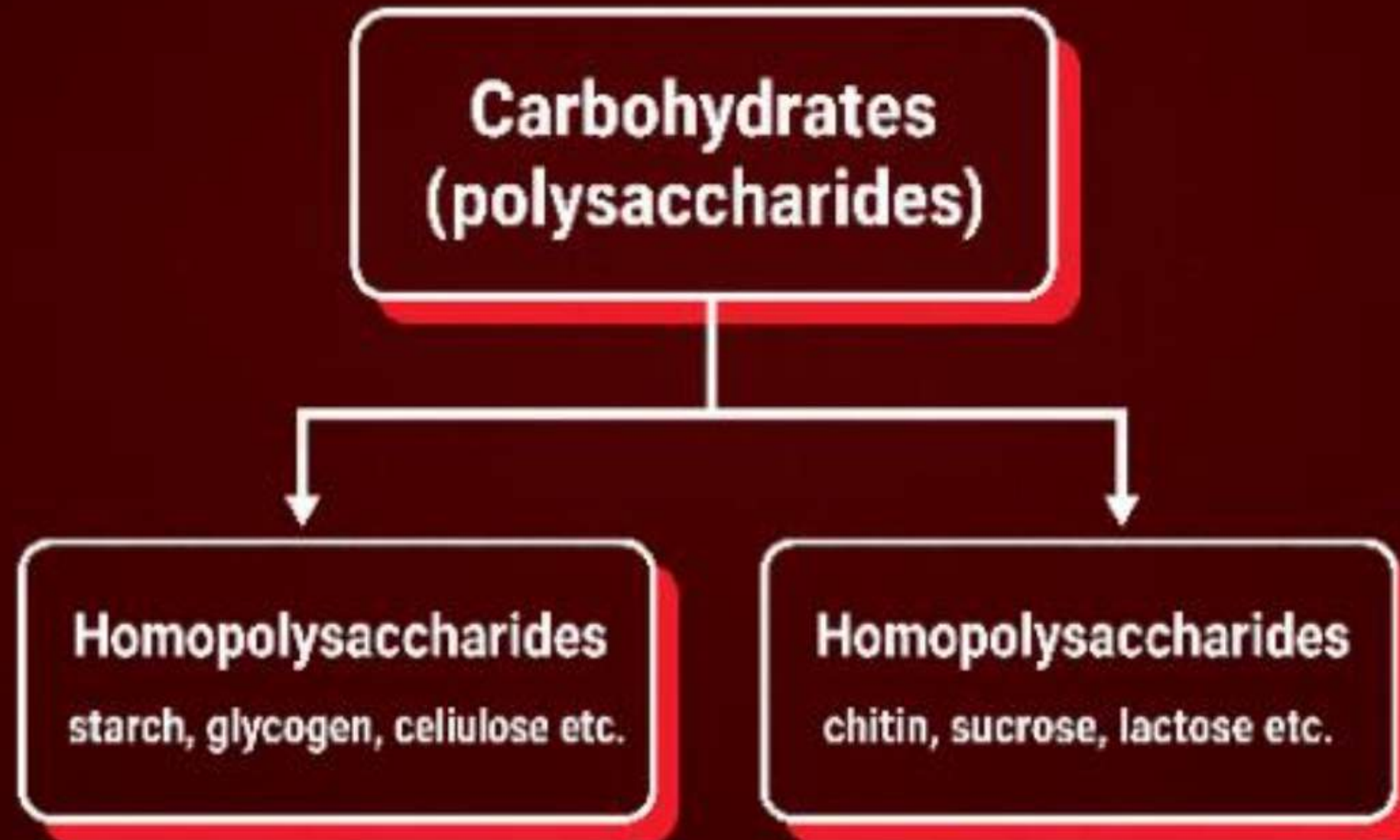
- ❑ In nucleic acids, the phosphate molecules links 3' C of sugar of one nucleoside to the 5' C of sugar of next nucleoside releasing two water molecules to form 3'-5' **phosphodiester bond**.
- ❑ In polysaccharides, the mono-saccharides are linked by **glycosidic bonds** formed by dehydration between two carbon atoms of two adjacent monosaccharides.





Carbohydrates (Polysaccharides)

- Polysaccharides are long chain of sugar containing different monosaccharaides as a building block.





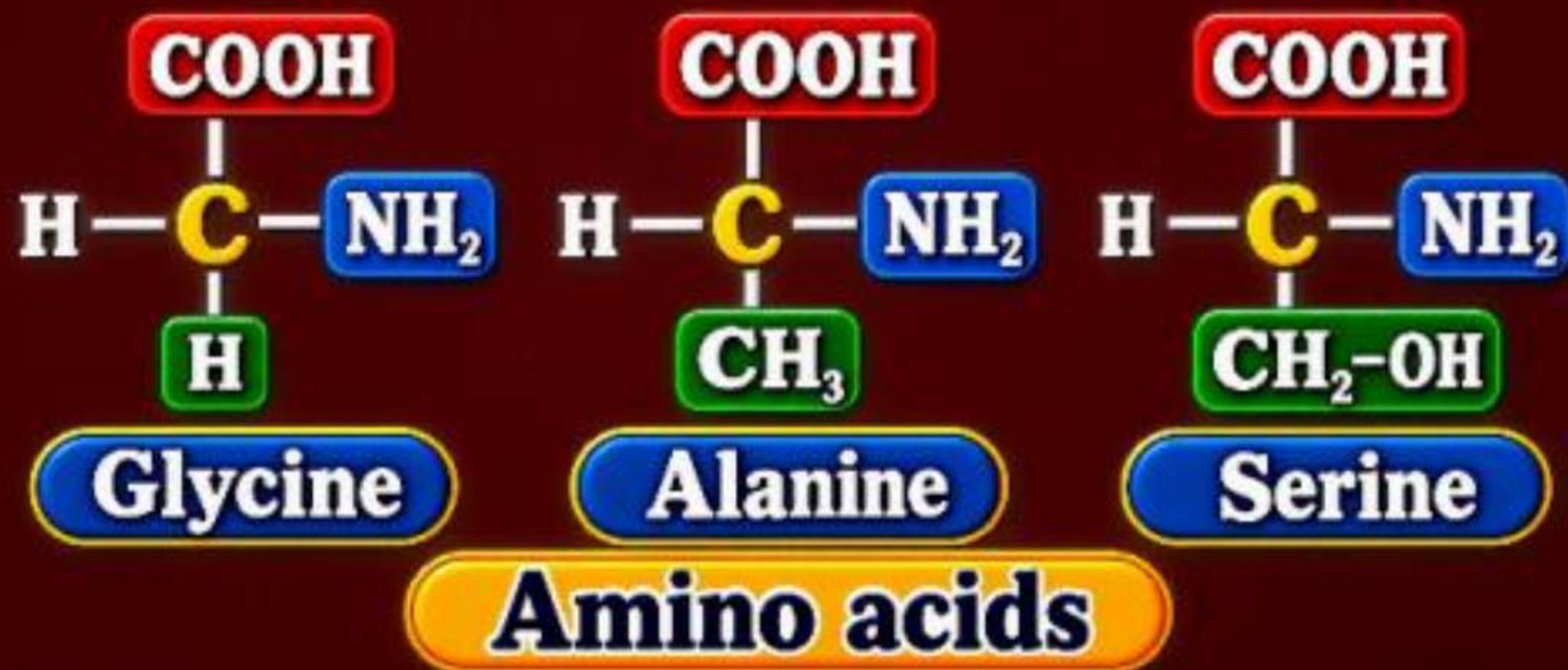
Carbohydrates (Polysaccharides) (Continued)

- * Starch is present in plants as store house of energy. It forms helical secondary structure. It can hold the I_2 molecules in the helical structure.
- * Cellulose molecules contain glucose molecules joined together by 1-4 β linkage. It is the most abundant organic molecules on earth.
- * Glycogen is called animal starch as it is the reserve food materials for animals, bacteria and fungi. In this, glucose molecules are arranged in highly branched bush like chain having two types of linkage 1-4 α in straight chain and 1-6 linkage in branching.
- * **Proteins** are polypeptide chains made up of amino acids. There are 20 types of amino acids joined together by peptide bond between amino and carboxylic group. There are two kinds of amino acids–



Types of Amino Acids

1. **Essential amino acids** are obtained by living organism along with food.
2. **Non-essential amino acids** can be prepared by our body from raw materials.





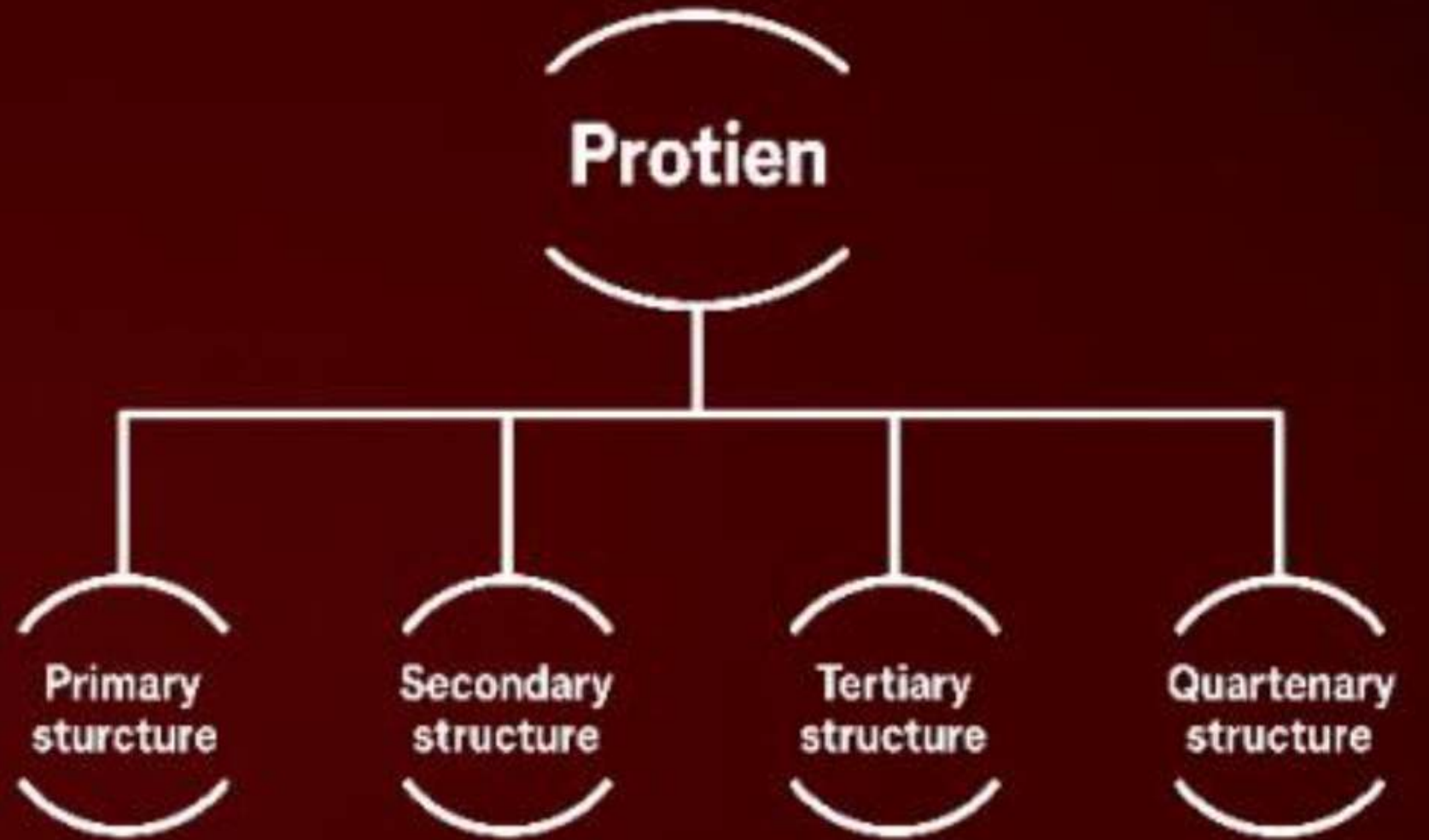
☒ **The main functions of protein in living cell are**

1. Transport of nutrient across the membrane.
2. Fight infectious organisms.
3. Produce enzyme and proteins.

Collagen is the most abundant protein in animal world.



- ▣ **Primary structure of protein** is the basic structure of protein in which a number of polypeptides are involved having sequence of amino acids. The first amino acid of sequence is called N-terminal amino acid and last amino acid of peptide chain is called C-terminal amino acid.

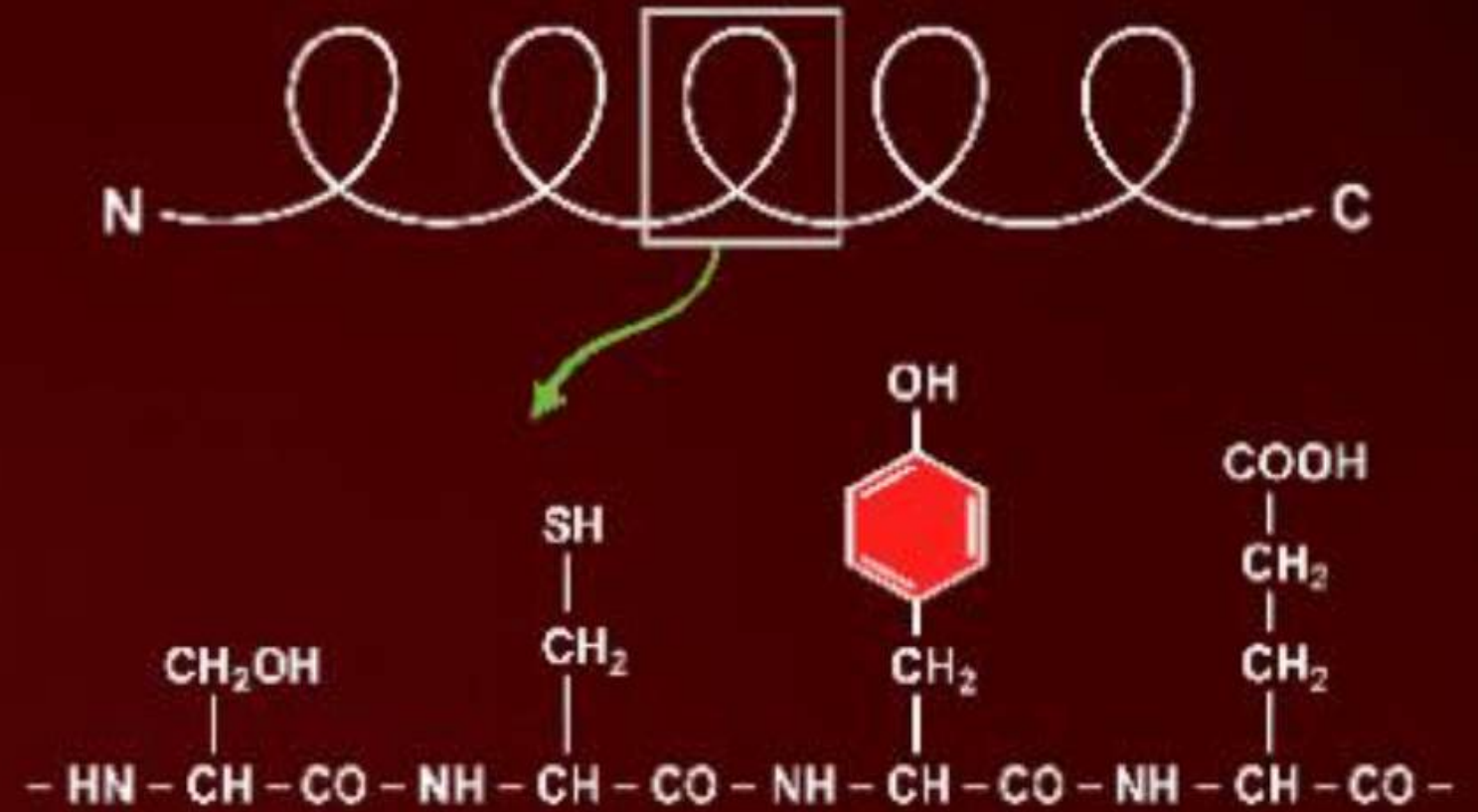




Secondary Structure of Protein



- ▣ **Secondary structure protein** threads forms helix. There are three types of secondary structure- α helix, β pleated and collagen. In a **helix**, the polypeptide chain is coiled spirally in right handed manner.
- ▣ In **β pleated** secondary proteins two or more polypeptide chains are interconnected by hydrogen bonds. In **collagen** there are three strands or polypeptides coiled around one another by hydrogen bonds.



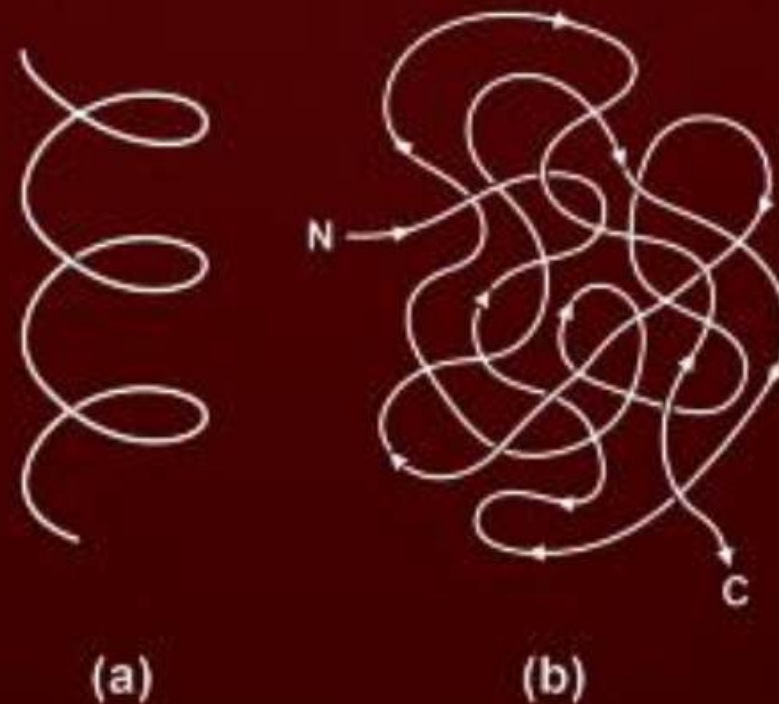


Tertiary and Quaternary Structure of Protein

- ☒ In **Tertiary structure** long protein chain is folded upon itself like a hollow woollen ball to give three dimensional view of protein.

(a) secondary structure (b) Tertiary structure

- ☒ In **Quaternary structure** each polypeptide develops its own tertiary structure and function as subunit of protein. Eg. Hemoglobin. In adult human haemoglobin 4 subunits are involved. The two subunits are of α type and two subunits of β types.





Enzymes

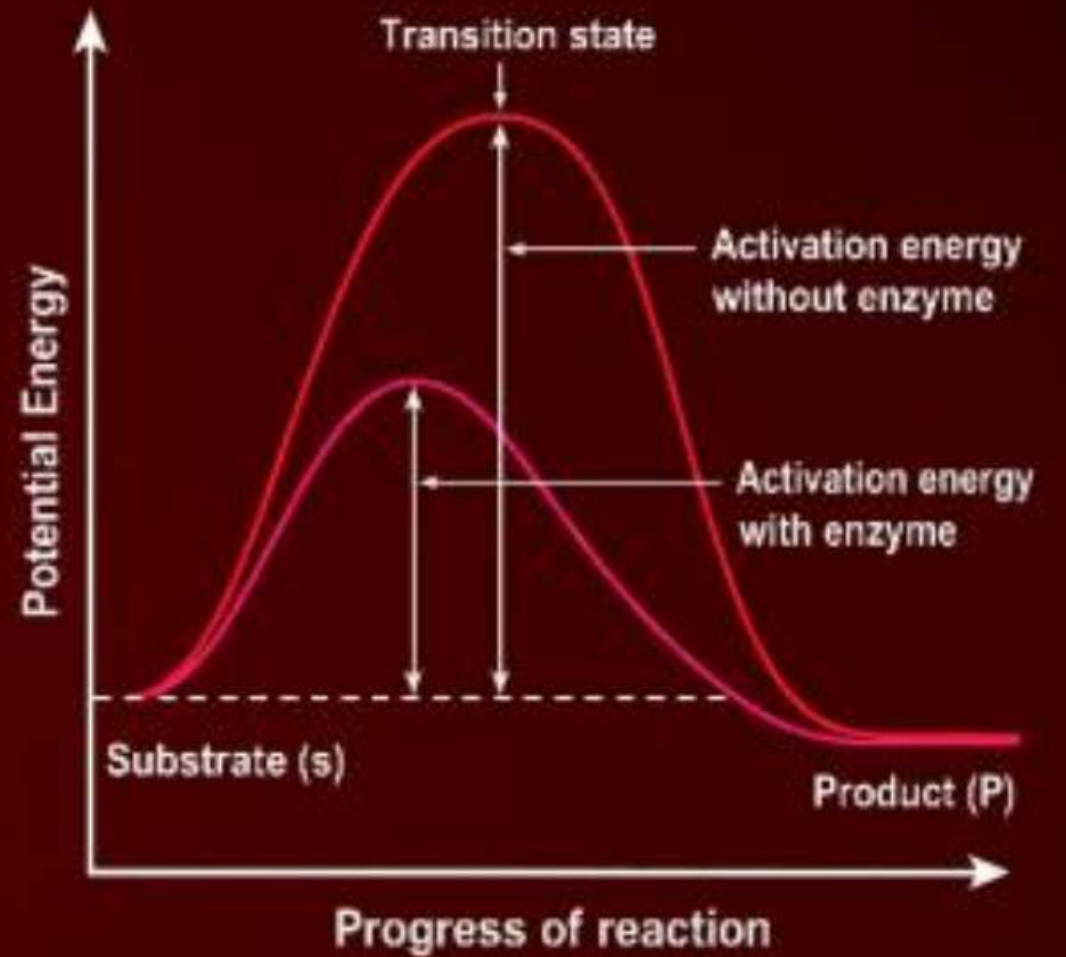


- ☒ Enzymes are commonly proteinaceous substances which are capable of catalysing chemical reactions of biological origin without themselves undergoing any change. They are commonly called as **biocatalysts**.
- ☒ The nucleic acids that behave like enzymes are called ribozymes.



Enzyme-Substrate (ES) Complex

- * The tertiary structure of protein/Enzyme has pockets or crevice into which substrate fit to form **ES complex**.
- * The formation of the ES complex is essential for catalysis.
- * $E + S \rightleftharpoons ES \rightarrow EP \rightarrow E + P$
- * The structure of substrate gets transformed into the structure of product through formation of transient state structure.
- * The major difference between inorganic and organic catalyst is inorganic catalyst works effectively at high temperature and pressure but enzyme get damaged at high temperature.
- * The external energy required to start a chemical reaction is called activation energy.

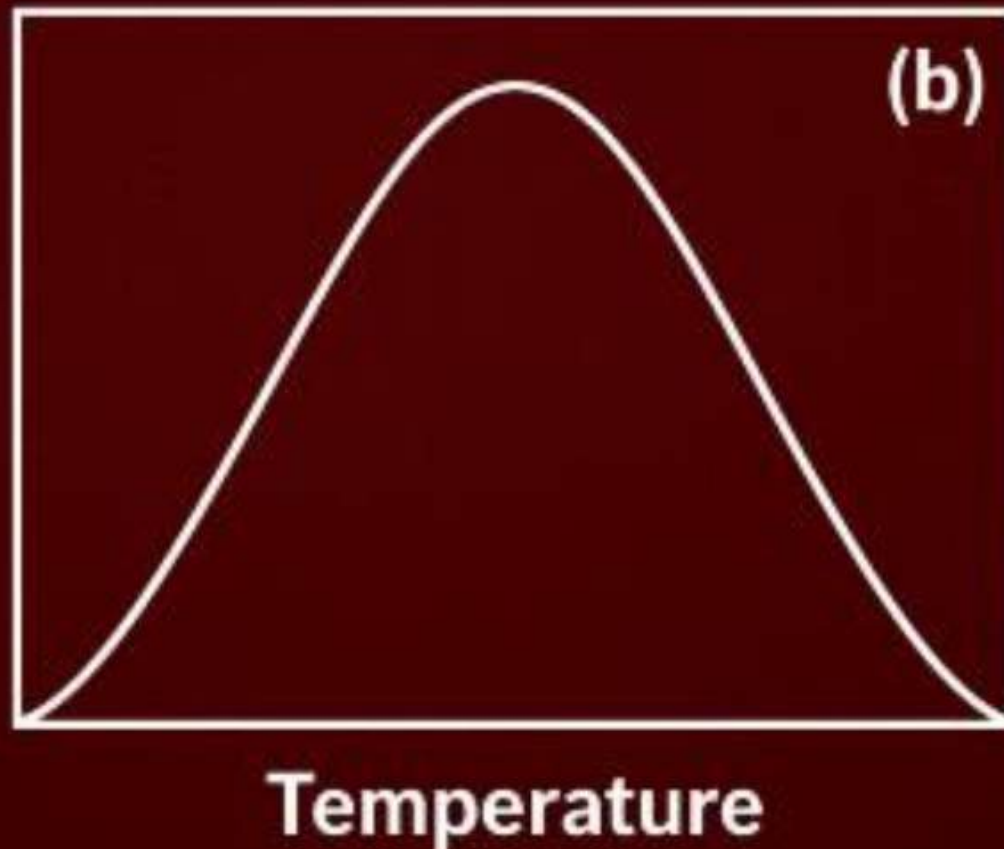




Factors influencing Enzyme Activity



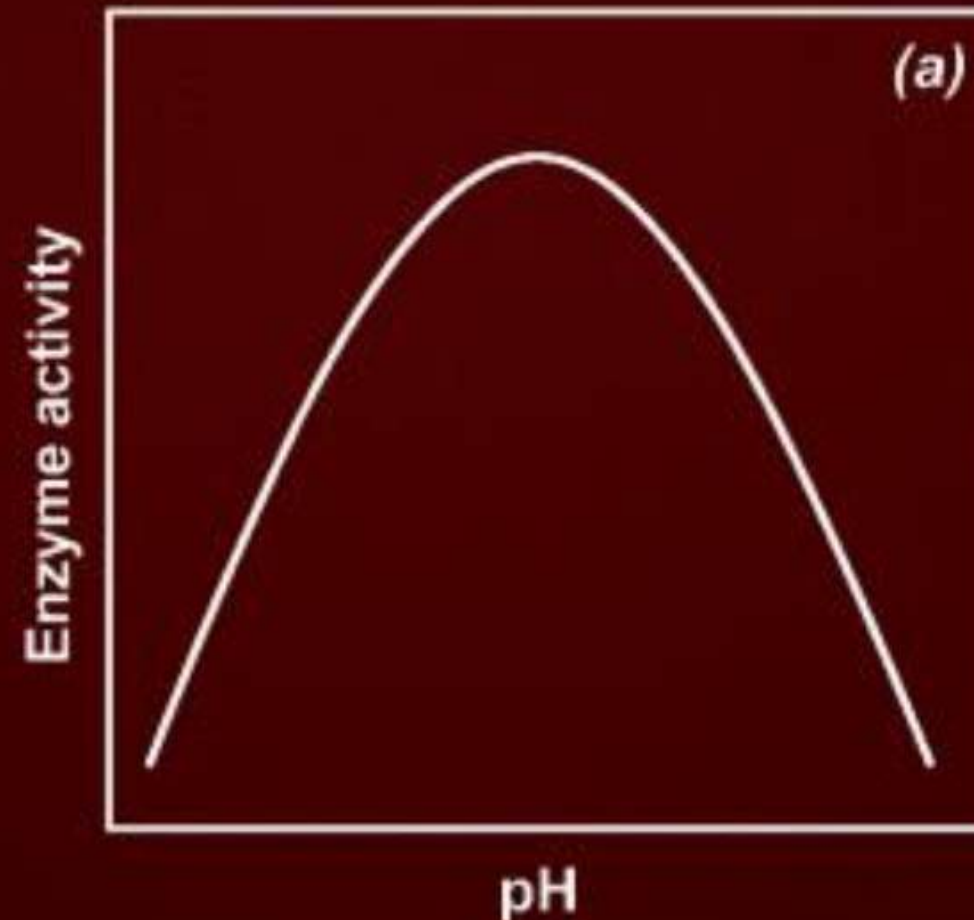
1. **Temperature**- An enzyme is active within a narrow range of temperature. Temperature at which enzyme is most active is called **optimum temperature**. The enzyme activity decreases above and below this temperature.





Factors influencing Enzyme Activity (Continued)

2. **pH** – every enzymes has an optimum pH at which it is maximum active. Most of the intracellular enzymes work at neutral pH.
3. **Concentration of Substrate**- increase in substrate concentration increases the rate of reaction due to occupation of more active sites by substrate.

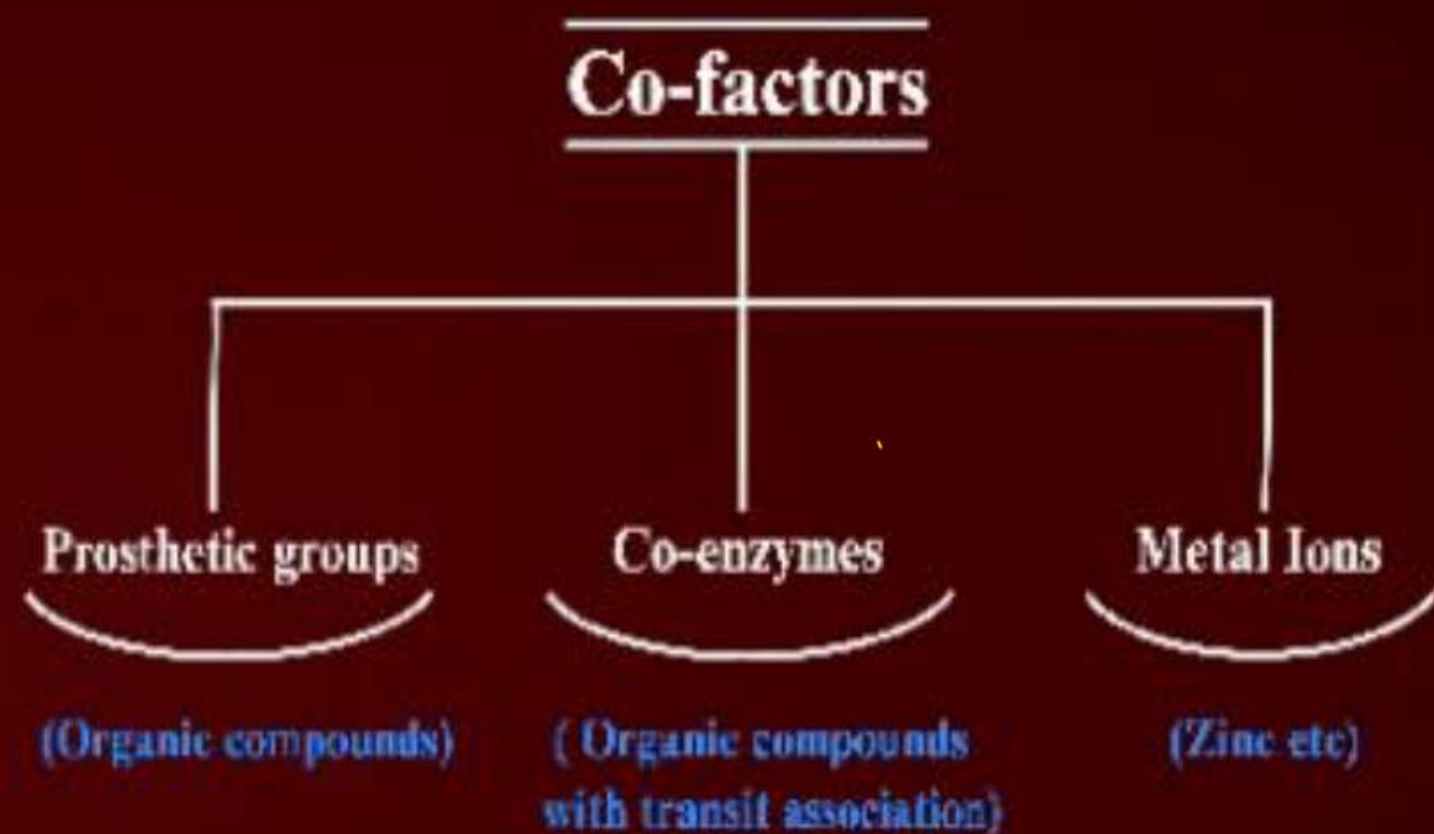




Co-factors



- ❑ **Co-factors** are the non-protein constituent of an enzyme which make the enzyme more catalytically active. The protein portions of enzyme are called apoenzyme.
- ❑ Prosthetic groups are organic compounds and are tightly bound to the apoenzyme. For example, in peroxidase and catalase, which catalyze the breakdown of hydrogen peroxide, haem is the prosthetic group
- ❑ The essential chemical components of any coenzymes are vitamins. Example, coenzyme NAD and NADP contain the vitamin niacin



Thank
You